

ME-203A: Kinematics of Machines

by

Dr. Emarti Kumari



**Department of Mechanical Engineering
M.B.M. Engineering College
Jai Narain Vyas University, Jodhpur**

SYLLABUS:

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

UNIT 2

Special purpose mechanism: Mechanisms for straight line, Motorcar steering mechanisms, Hooks joint and Geneva mechanism

UNIT 3

Kinematics synthesis of mechanism: Type of synthesis, DOF, Grashof's criterion of movability, Limit position, Dead center transmission angle synthesis of mechanism by relative pole method, Inversion method, Freudenstein's methods of synthesis, Design of a four bar chain for constant angular velocity ratio of cranks.

UNIT 4

Friction: Solid, rolling greasy and viscous friction, Laws of Friction inclined plane, Friction of pivots and collars, Friction circle of turning pair and friction axis of a link, Ball and roller bearings, Single and multiple disc-clutches

UNIT 5

Belts, Ropes & Chains: Mechanics of belt drive, Length of belt, Ratio of tensions in flat and V belt, Effect of centrifugal force, Power transmission capacity, Belt materials.

Rope drive: Types of ropes, Construction, Forces on rope and breaking strength,

Chain drive: Types, Chain length, Chordal action hoisting and power transmission

ME 203 A: KINEMATICS OF MACHINES

2L, 1T

3 Hours, 100 Marks

COURSE OUTCOMES:

CO1: Describe the concepts of mechanisms, machines and related terminologies and discuss different types of methods to analyse the velocity and acceleration of various mechanism.

CO2: Discuss Geneva mechanism.

CO3: Design of a four bar chain for constant angular velocity ratio of cranks.

CO4: Explain the concepts of friction and its types. In addition, assess its effects of different mechanical components.

CO5: Analyze various motion transmission elements like belt drive, rope drive and chain drive.

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

Reference Books:

1. Theory of Machines, Kinematics and Dynamics, By Sadhu Singh, Published by Pearson.
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3. Theory of Machines, By S. S. Rattan, Published by Mc Graw Hill Education.
4. Theory of Machine, R.S. Khurmi and J.K. Gupta, Published by Eurasia Publishing House (Pvt.) Ltd.

Definition of Kinematics:

- ❖ **Study of motion** of particles and rigid bodies without reference to forces.

Mechanisms and Machines:

- ❖ **Kinematics of inter-connected rigid bodies.**

Equilibrating forces:

- ❖ **quasi-static analysis.**

Mechanism:

Combination of rigid bodies (links) so shaped and connected that they move relative to each other for the purpose of transferring and transforming motion / forces from input (s) (actuator (s)) to an output.

Machine:

A device that uses mechanism (s) to transform and transfer motion and force from a power source to perform some useful mechanical work.

Mechanisms and Machines:

- ❖ Transmission and transformation of motion and forces.

Motion:

- ❖ Displacement, velocity, acceleration, path.

Forces transmission: actuator forces

Motivation:

- **Mechanization and Automation:** requirement of specialized mechanical devices (manufacturing, mechanical handling, assembly, painting, packaging).
- **Health-care:** transfer aids and devices, physiotherapy surgery.
- **Synthesis and design:** requires kinematic analysis.

Link or Element:

- ❖ Each part of a machine which has motion relative to some other part is termed as element or link.
- ❖ A link need not necessarily be a rigid body, but it must be a resistant body. It must be capable of transmitting the required force with negligible deformation.

Examples: links which are resistant, but not rigid.

1. Liquids, which are resistant to compressive forces and are used as links in hydraulic presses, i.e. hydraulic brakes and hydraulic jacks.
2. Chains, belts and ropes; which are resistant to tensile forces and are used for transmitting motion and forces.

Classification of Links:

1. Binary link
2. Ternary link
3. Quaternary link

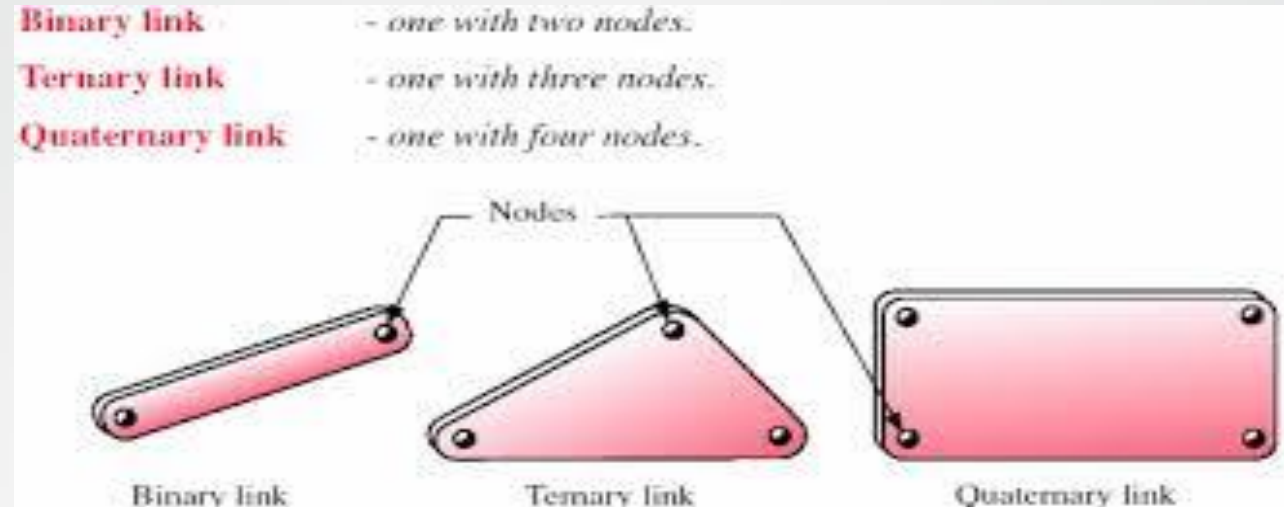


Fig. Reference mechediploma.com

There are four types of links:

1. **Rigid link**: a rigid link does not undergo any deformation while transmitting motion. Links in general are elastic in nature. E.g. connecting rod, crank, etc.
2. **Flexible link**: A flexible link is one which while transmitting motion is partly deformed in a manner not to effect the transmission of motion. E.g. belts, ropes, chains, springs, etc.
3. **Fluid link**: A fluid link is deformed by having fluid in a closed vessel and the motion is transmitted through the fluid by pressure. E.g. hydraulic press, hydraulic jack and fluid brake.
4. **Floating link**: It is a link which is not connected to the frame.

Kinematic Pair:

- ❖ The two links of a machine, when in contact with each other, are said to form a pair.
- ❖ Two elements or links which are connected together in such a way that their relative motion is completely constrained from a kinematic pair.

Classification of Kinematic Pairs:

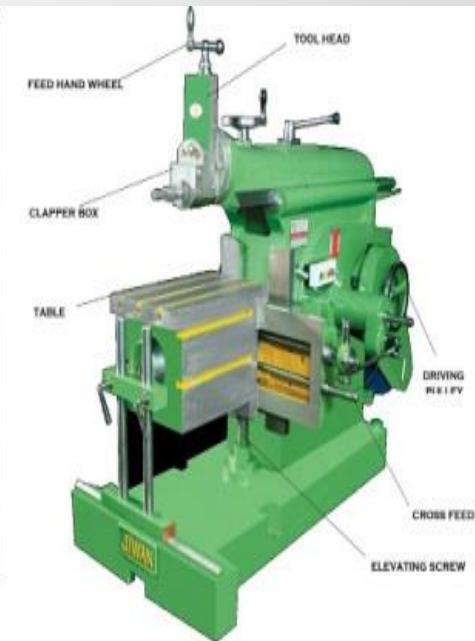
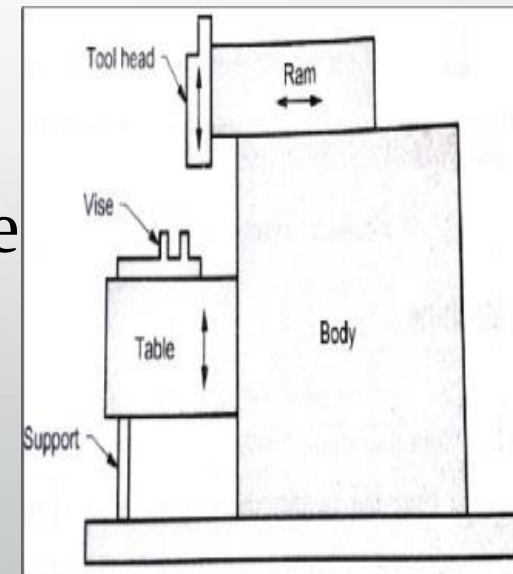
1. Type of relative motion
2. Type of contact
3. Type of mechanical constraint

1. Kinematic pairs according to the relative motion:

- **Sliding pair:** It consists of two elements connected in such a manner that one is constrained to have sliding motion to another.

For example:

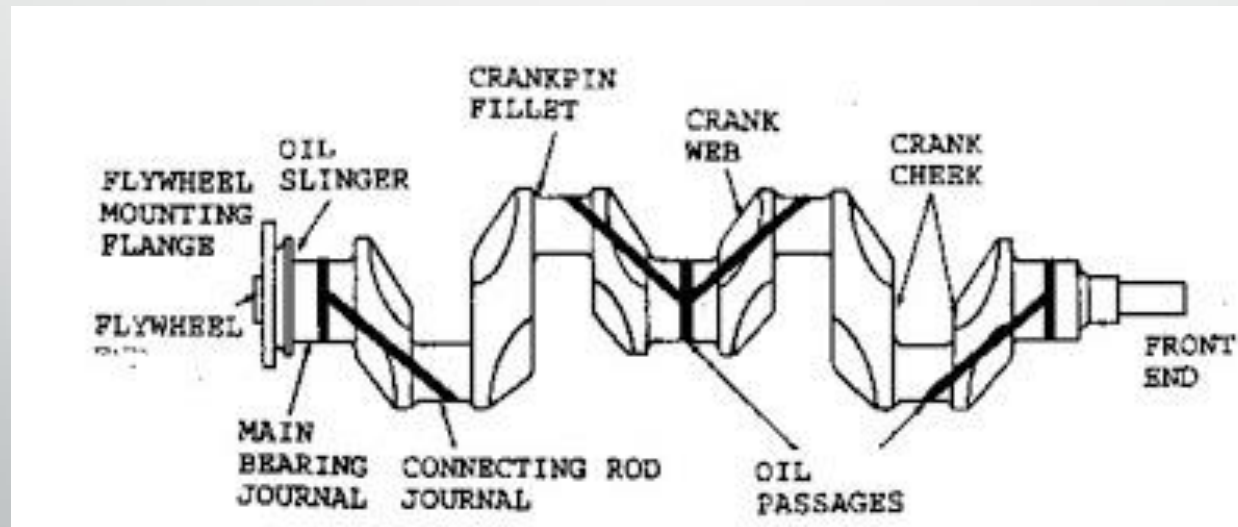
1. rectangular bar in rectangular hole
2. Piston and cylinder of an engine
3. Cross-head and guides of a steam engine
4. Ram and its guides in a shaper
5. Tailstock on the lathe bed



- **Turning (revolute) pair:** It consists of two elements connected in such a manner that one is constrained to turn or revolve about a fixed axis of another element.

For example:

1. a shaft with collar at both ends revolving in a circular hole
2. Crankshaft turning in a bearing
3. Cycle wheels revolving over their axles.



- **Roller pair:** when two element are connected that one is constrained to roll on another element which is fixed, form a rolling pair.

For example:

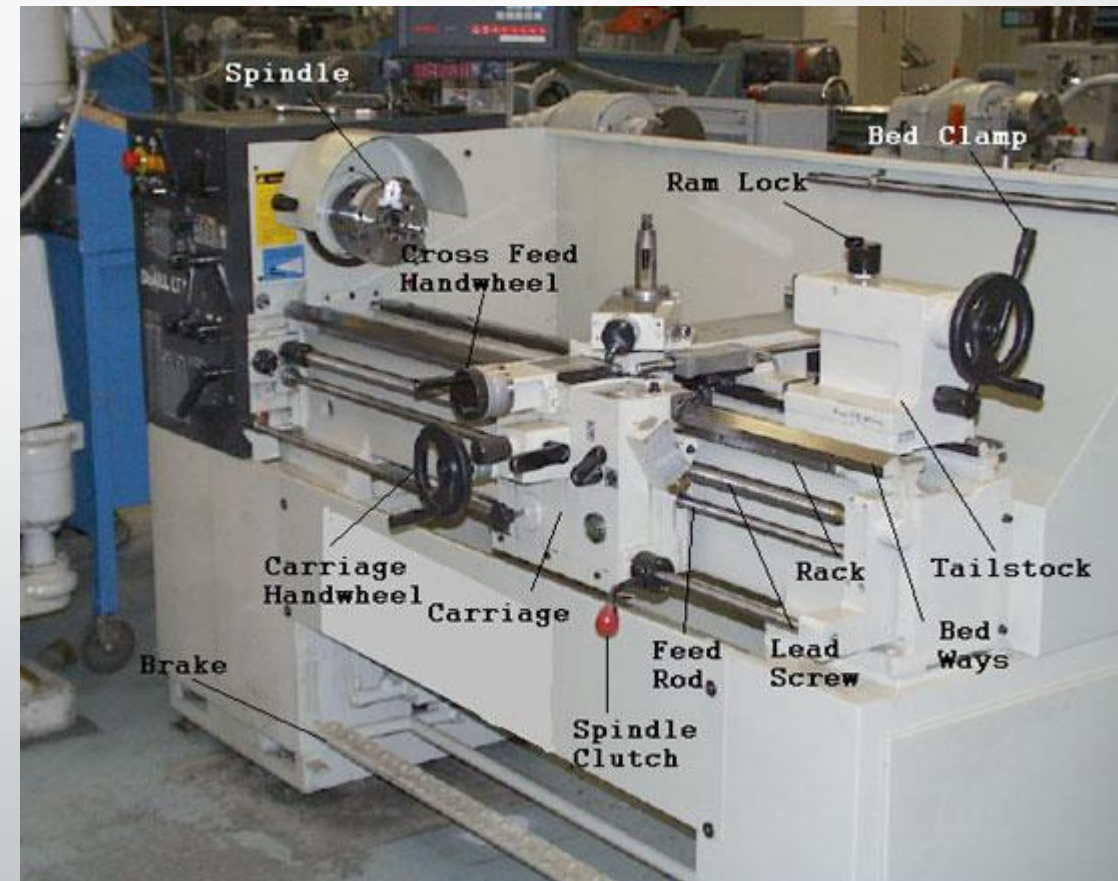
1. Ball and roller bearing
2. A wheel rolling on a flat surface



- **Screw (or helical) pair:** when one element turns about the other element by means of threads, it forms a pair. The motion in this case is a combination of sliding and turning.

For example:

1. Lead screw of a lathe machine with nut
2. Screw with nut of a jack
3. Bolt with nut



- **Spherical pair:** when one element in the form of a sphere turns about the other element, it forms a spherical pair.

For example:

1. Ball and socket joint
2. Pen stand
3. The mirror attachment of vehicles

Ball & Socket Joint
eg. Hip Joint

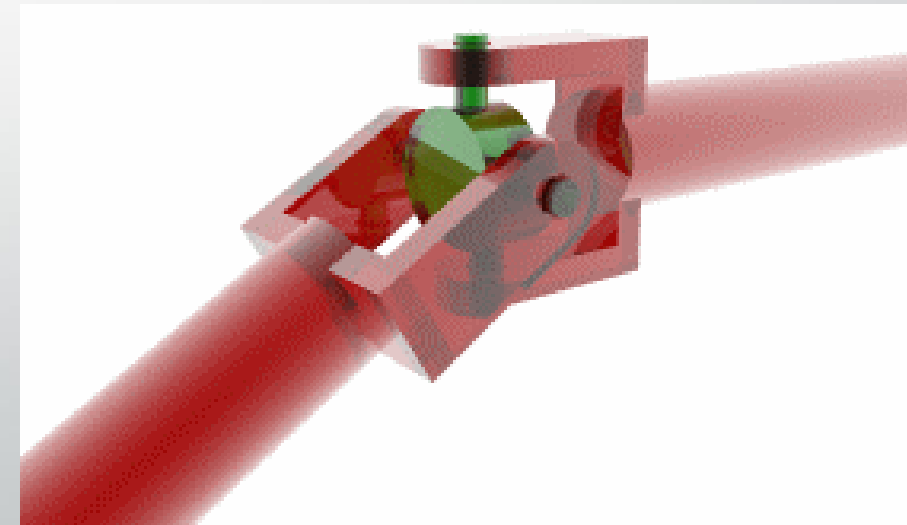


2. Kinematic pairs according to the type of contact:

- **Lower pair:** when two elements have surface (area) contact while in motion and relative motion is purely turning or sliding, they are called a lower pair.
- All the sliding pair, turning pair, and screw pair form a lower pair.

For example:

1. Nut turning on a screw
2. Shaft rotating in a bearing
3. Universal joint
4. All pairs of slider crank mechanism
5. Pantograph

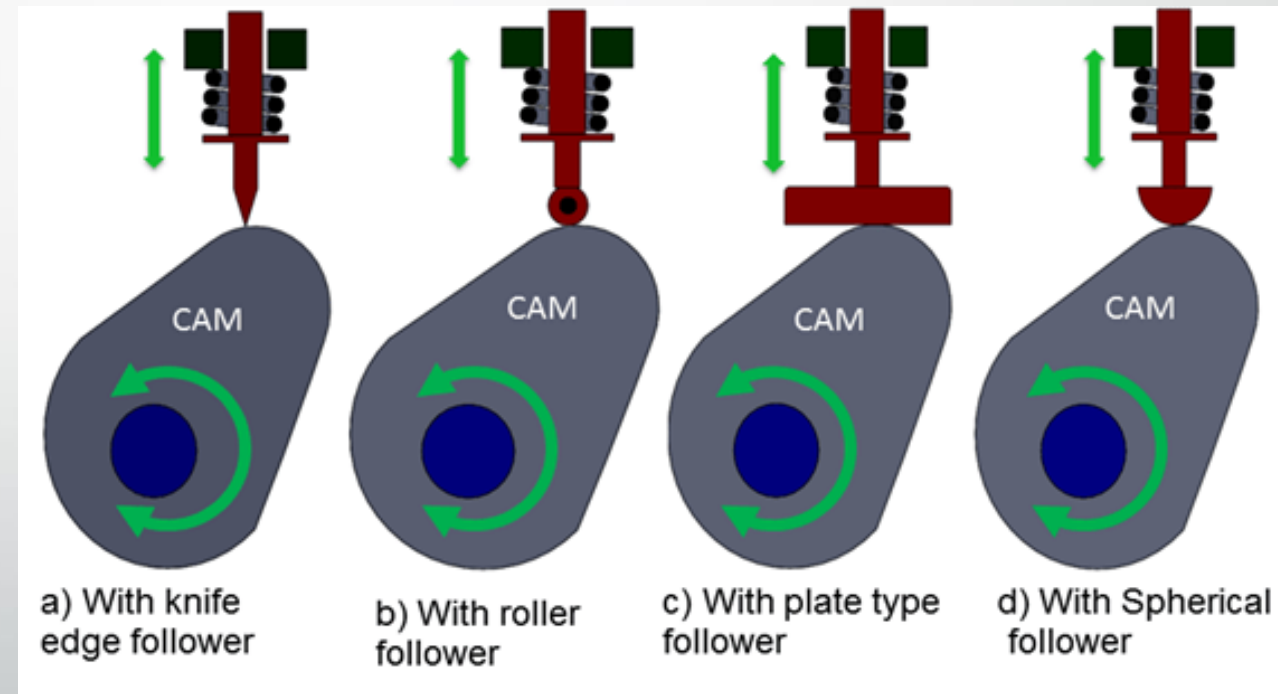


Reference: youtube.com

- **Higher pair:** when two elements have point or line contact while in motion and relative motion is purely turning or sliding, then the pair is known as higher pair.

For example:

1. Belts
2. Ropes
3. Chain drive
4. Gears
5. Cam and follower
6. Ball and roller bearings
7. Wheel rolling on a surface



3. Kinematic pairs according to Mechanical constraint:

- **Closed pair:** when two elements of a pair held together mechanically in such a manner that only the required type of relative motion occurs, they are called closed pair.

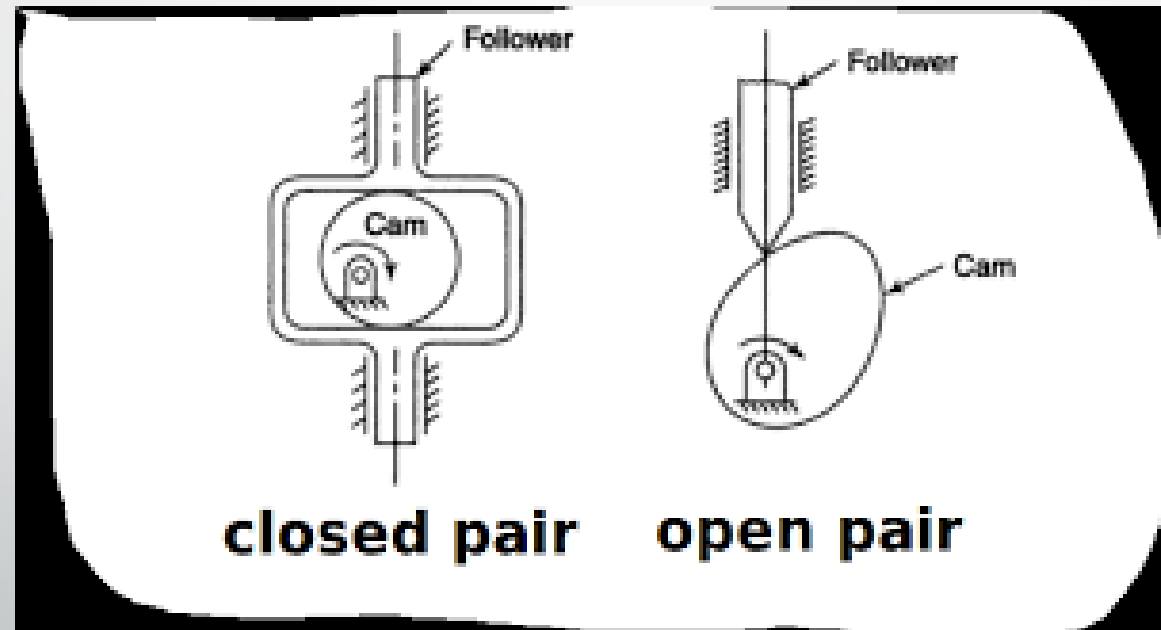
For example:

1. All the lower pairs and some of higher pair (enclosed cam and follower) are closed pairs.

- **Unclosed pair:** when two elements of a pair are not held mechanically and are held in contact by the action of external forces, are called unclosed pair.

For example:

1. Cam and spring loaded follower pair

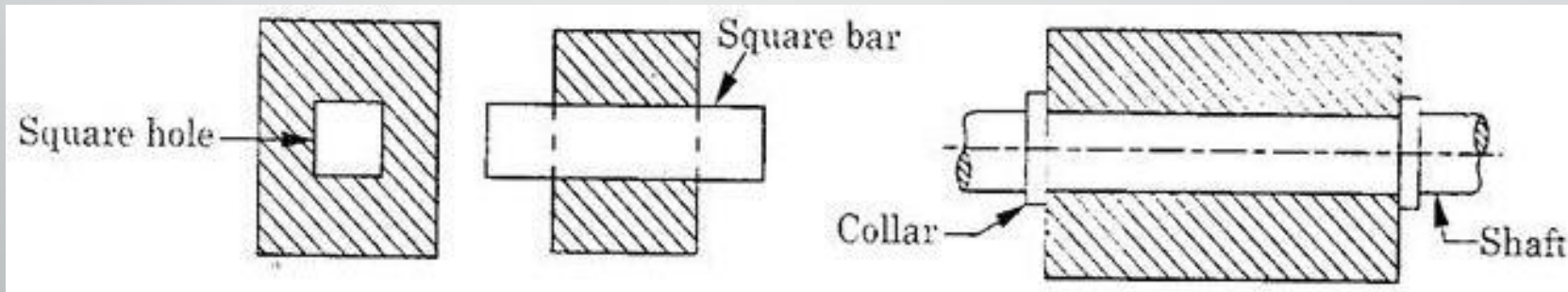


Constrained Motion

1. **Completely constraint motion:** when the motion between a pair takes place in a definite direction irrespective of the direction of force applied, then the motion is said to be a completely constrained motion.

For Example:

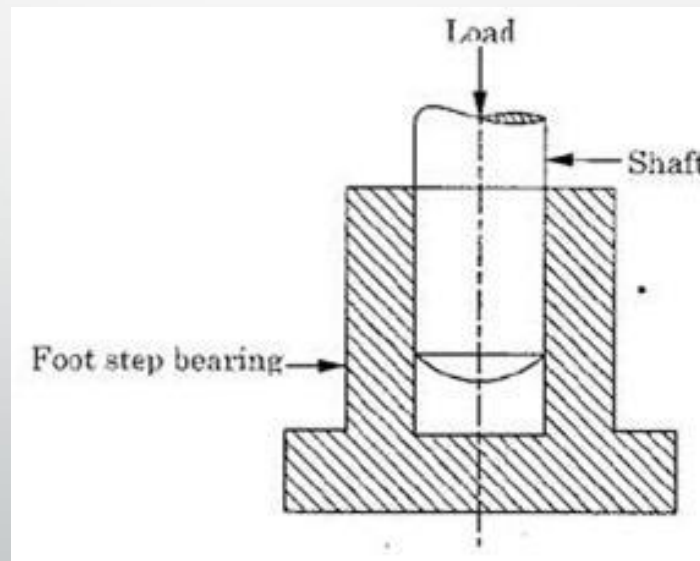
1. A square bar in square hole
2. A shaft with collars at each end in a circular hole
3. A piston in the cylinder of an internal combustion engine



2. Partially or successfully constrained motion: when the constrained motion between a pair is not completed itself but by some other means, then the motion is said to be a successfully constrained motion.

For Example:

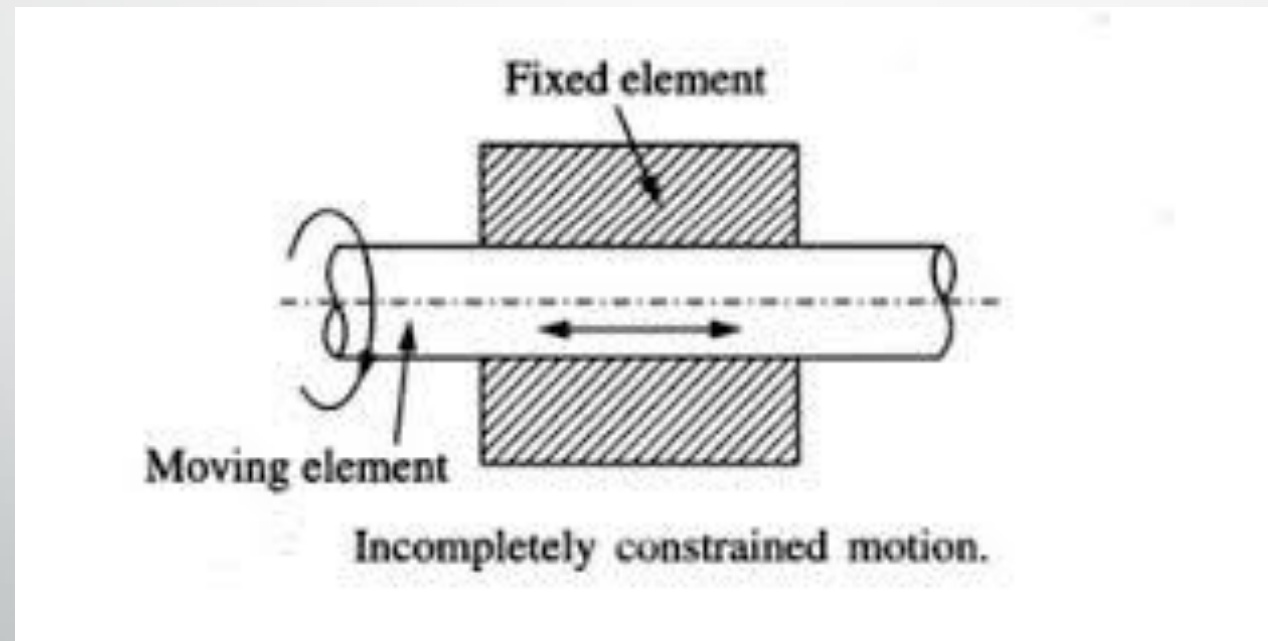
1. The motion of a shaft in a footstep bearing becomes successfully constrained motion when compressive load is applied to the shaft.



3. Incompletely constrained motion: when the motion between a pair can take place in more than one direction, it is said to be incompletely constrained motion.

For Example:

1. A circular shaft in a circular hole.





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Kinematic Chain:

A kinematic chain may be defined as an assembly of links in which the relative motion of the links is possible and the motion of each relative to the others is definite. The last link of the kinematic chain is attached to the first link. The four-bar mechanism and the slider crank mechanism are some of the examples of a kinematic chain.

➤ **Relationship for kinematic chain having lower pairs:**

$$L = 2P - 4$$

$$J = 3L/2 - 2$$

Where: L = number of binary links

P = number of lower pairs

J = number of binary joints

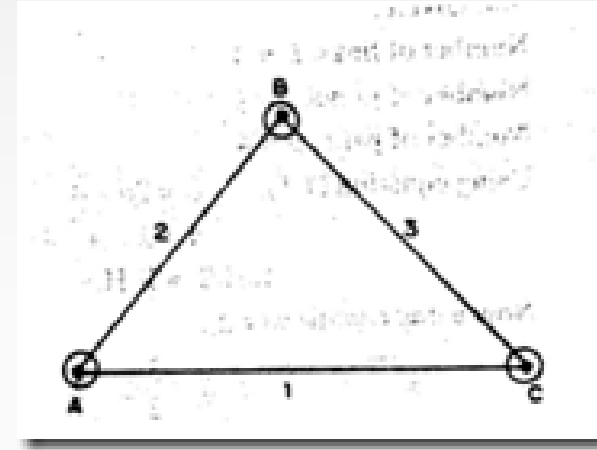
If $LHS > RHS$, then chain is called locked chain or redundant chain
$LHS = RHS$, then chain is constrained
$LHS < RHS$, then chain is unconstrained

- **For a kinematic chain having higher pairs, each higher pair is taken equivalent to two lower pairs:**

$$J + H/2 = 3L/2 - 2$$

Where H = number of higher pairs.

Example 1: A chain with three links is shown here.
Prove that the chain is locked.



Reference: researchgate.net

Solution: Number of binary joints, $J = 3$
Number of binary links, $L = 3$
Number of lower pairs, $P = 3$

Firstly, No. of lower links:

$$L = 2P - 4$$

$$3 = 2 \times 3 - 4 = 2$$

$$LHS > RHS$$

Secondly, No. of binary joints:

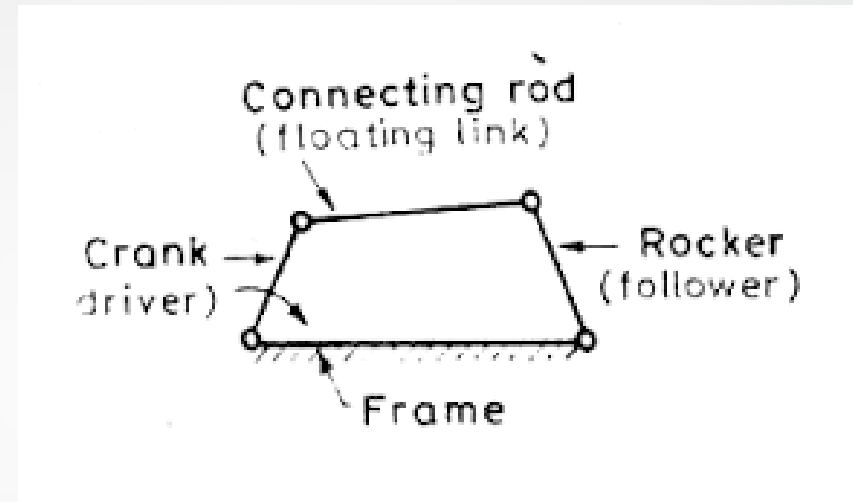
$$J = 3L/2 - 2$$

$$3 = 3 \times 3/2 - 2 = 2$$

$$LHS > RHS$$

Therefore, it is a locked chain.

Example 2: A four-bar chain is shown here. Prove that it is a constrained chain.



Reference: researchgate.net

Solution: Number of binary joints, $J = 4$
 Number of binary links, $L = 4$
 Number of lower pairs, $P = 4$

Firstly, No. of lower links:

$$L = 2P - 4$$

$$4 = 2 \times 4 - 4 = 4$$

$$\text{LHS} = \text{RHS}$$

Secondly, No. of binary joints:

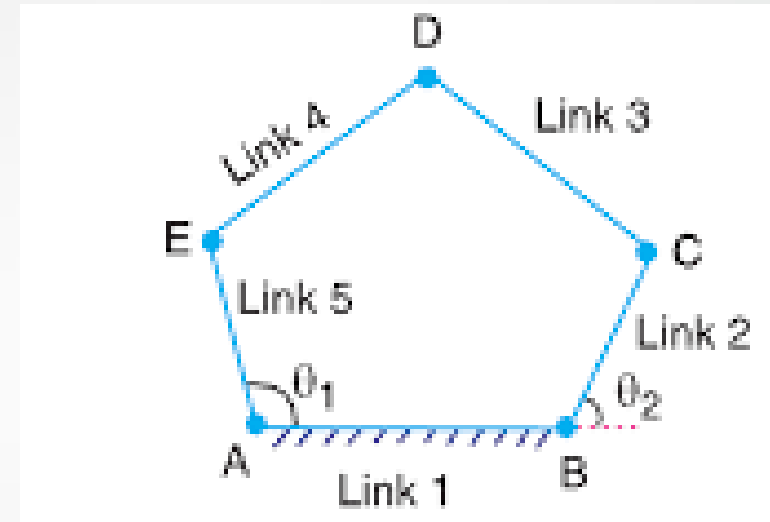
$$J = \frac{3L}{2} - 2$$

$$4 = \frac{3 \times 4}{2} - 2 = 4$$

$$\text{LHS} = \text{RHS}$$

Therefore, it is a constrained chain.

Example 3: A five-bar chain is shown here. Prove that it is an unconstrained chain.



Reference: researchgate.net

Solution: Number of binary joints, $J = 5$
 Number of binary links, $L = 5$
 Number of lower pairs, $P = 5$

Firstly, No. of lower links:

$$L = 2P - 4$$

$$5 = 2 \times 5 - 4 = 6$$

$$\text{LHS} < \text{RHS}$$

Secondly, No. of binary joints:

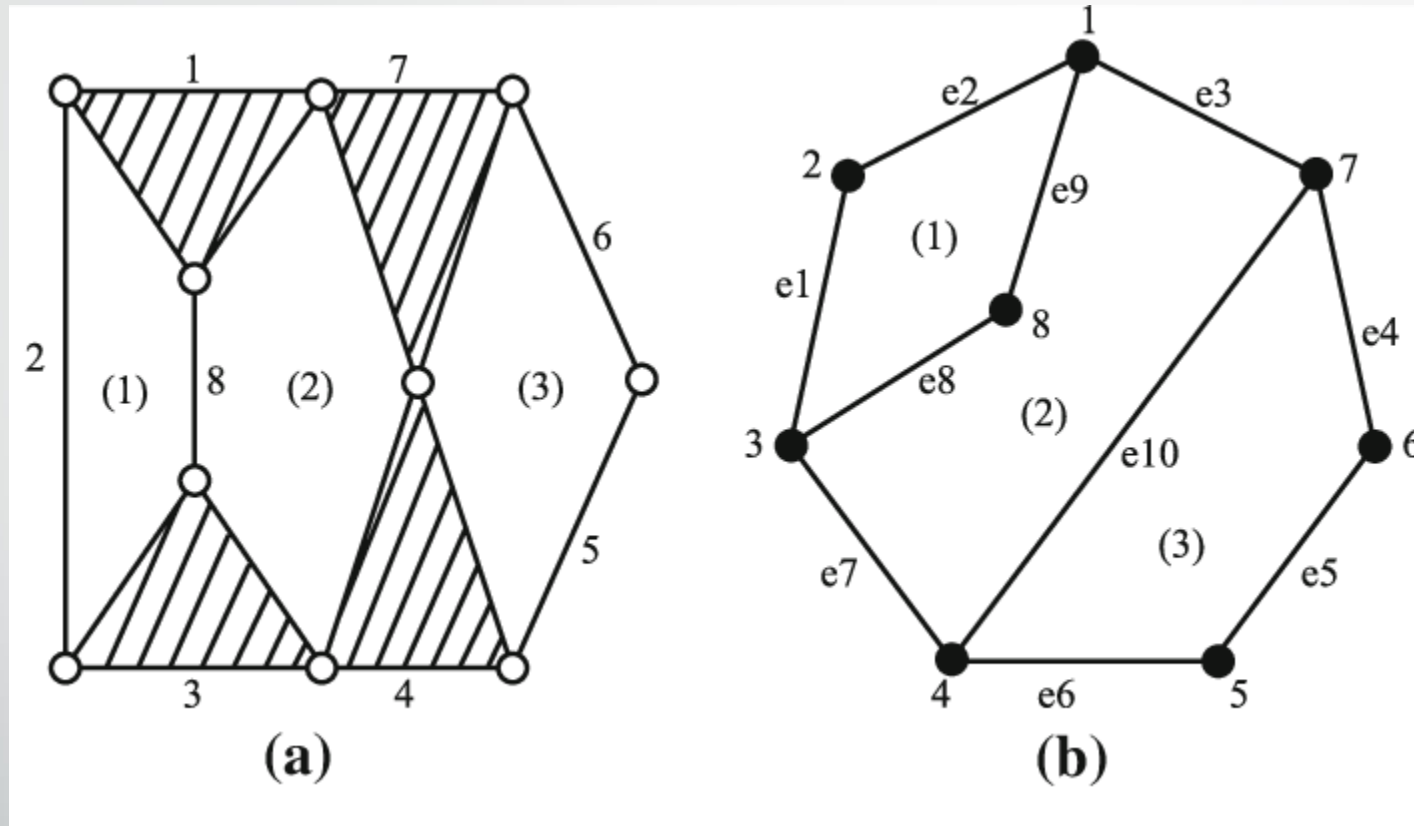
$$J = \frac{3L}{2} - 2$$

$$5 = \frac{3 \times 5}{2} - 2 = 5.5$$

$$\text{LHS} < \text{RHS}$$

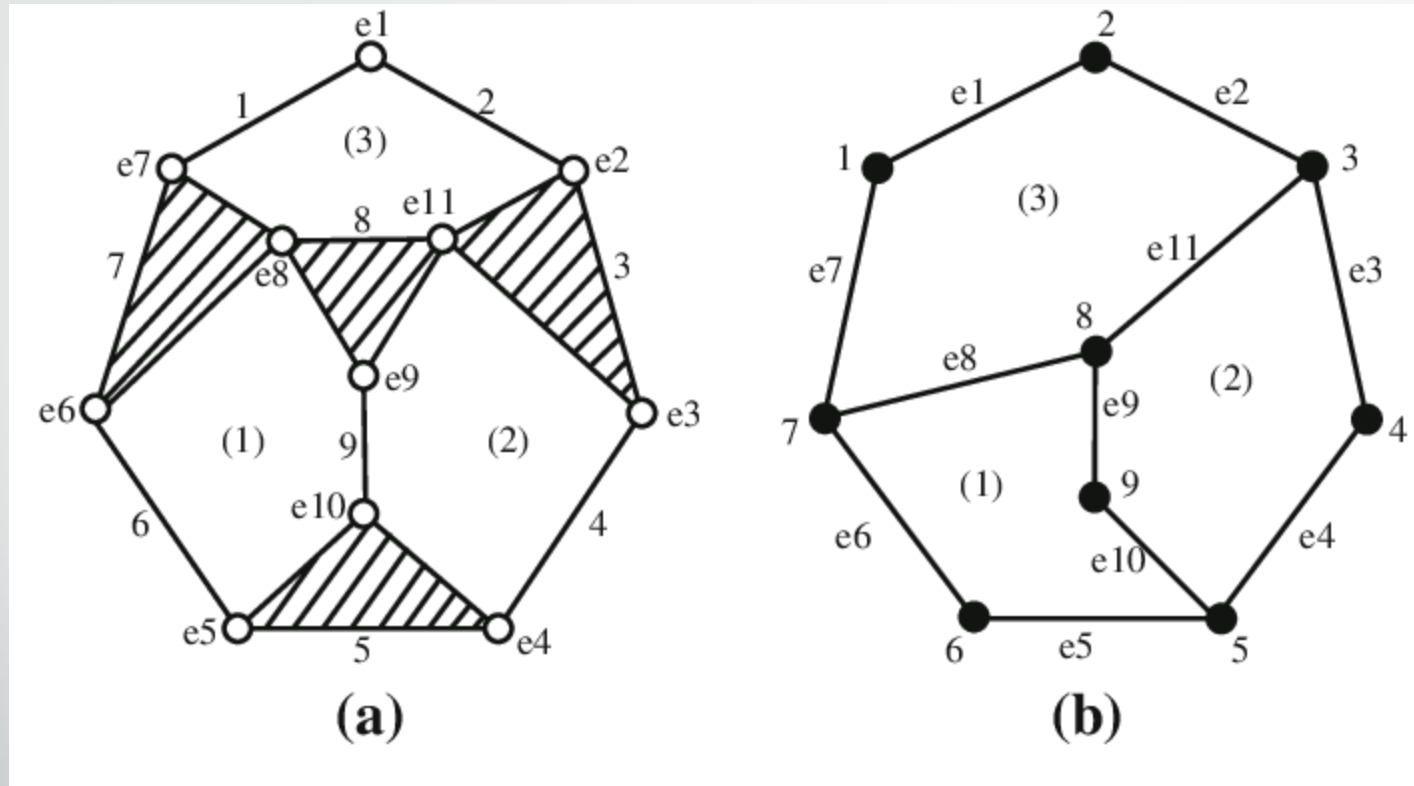
Therefore, it is an un-constrained chain.

Example 4: A eight-bar chain is shown here. Prove that it is an unconstrained chain.



Reference: researchgate.net

Example 5: A nine-bar chain is shown here. Prove that it is an unconstrained chain.



Reference: researchgate.net

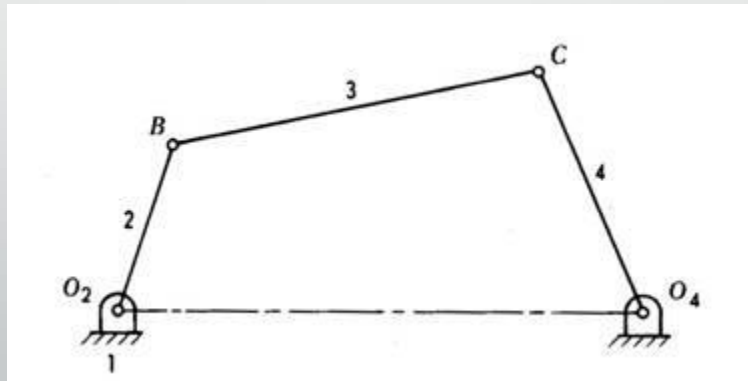
Mechanism:

When one of the links of a kinematic chain is fixed, the chain is called a mechanism.

Types of mechanisms:

Simple mechanism: a mechanism which has four links.

- In a simple machine, there is one point of application for the effort and one point for the load to be lifted.
- Simple machines are: lever, screw jack, inclined plane, bicycle.

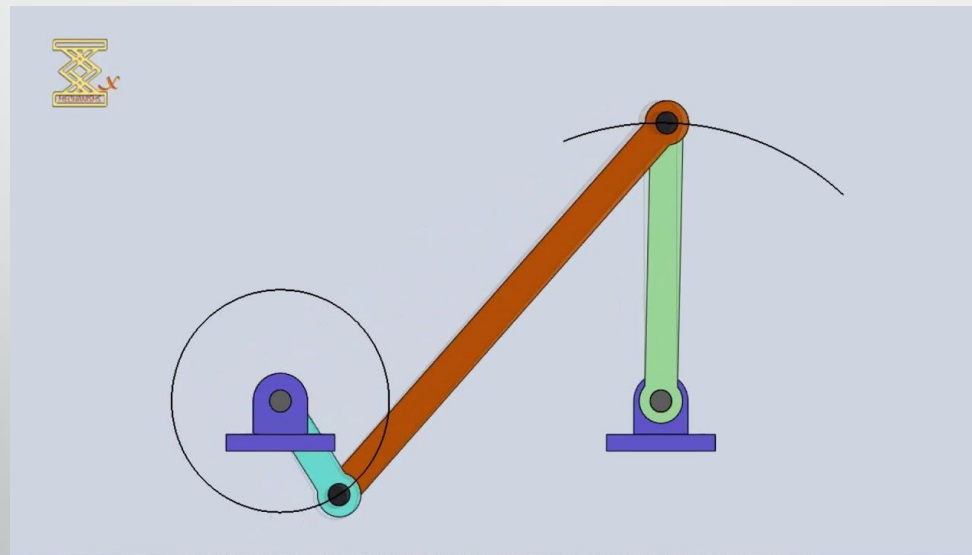


Reference: engineering.myindialist.com

Compound mechanism: a mechanism which has more than four links.

- In a compound machine, there are more than one point of application for the effort and the load.
- It may be thought of as a combination of many simple machines.

Compound machines are: Lathe machine, Grinding machine, Milling machine, Printing machine.

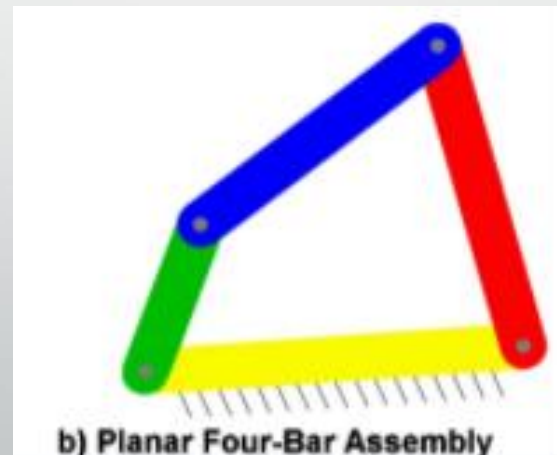


Complex mechanism: it is formed by the inclusion of ternary or higher order floating link to a simple mechanism.



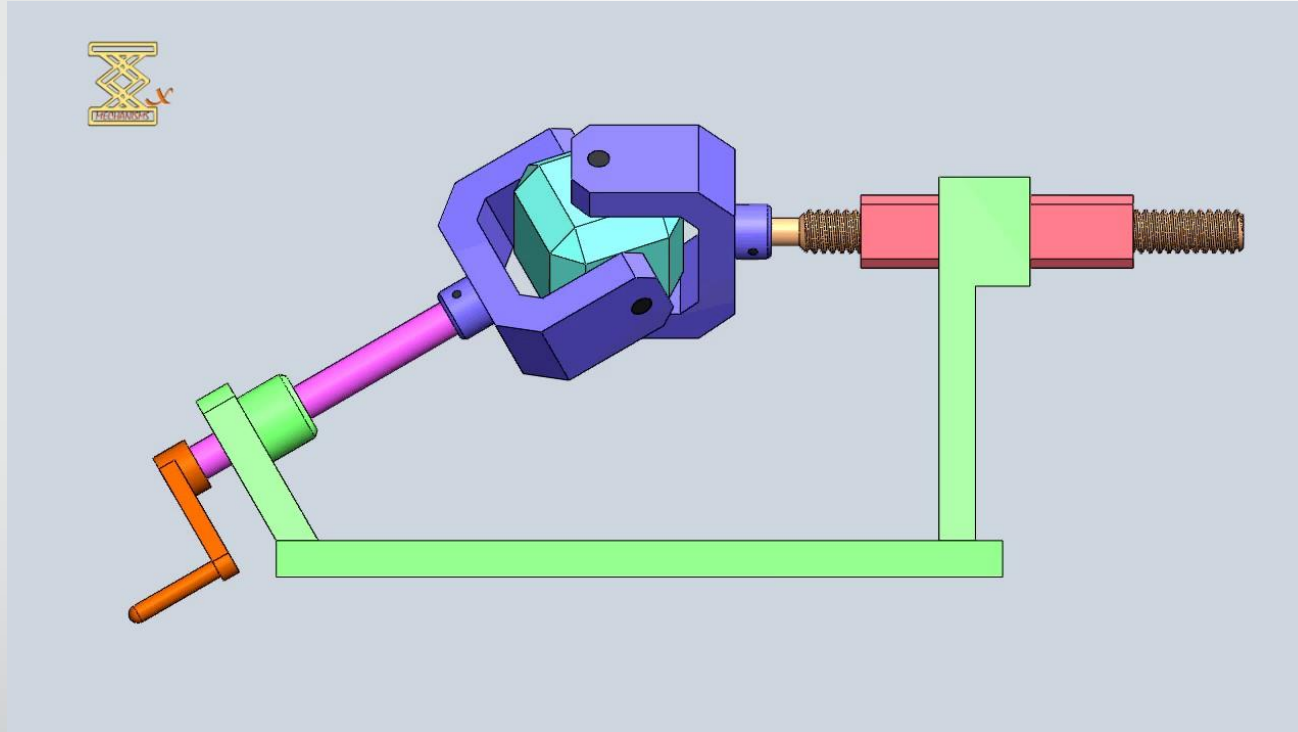
Reference: colourbox.com

Planar mechanism: when all the links of the mechanism lie in the same plane.



Reference: researchgate.net

Spatial mechanism: when the links of the mechanism lie in different planes.



Reference: youtube.com

Mechanism and Machines:

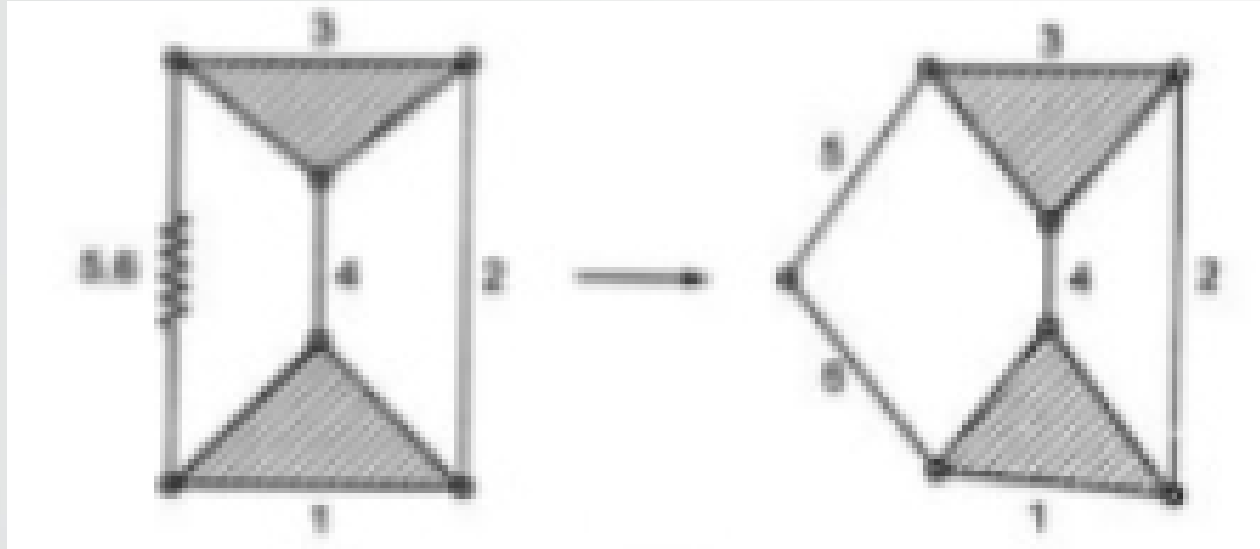
- A machine is a device that transforms energy available in one form to another to do certain type of desired useful work.
- Links may transmit both power and motion.
- A mechanism is a combination of rigid or restraining bodies, which are so shaped and connected that they move upon each other with definite relative motion.
- A mechanism is obtained when one of the links of the kinematic chain is fixed.
- A machine is a combination of two or more mechanisms arranged in such a way so as to obtain the required motion and transfer the energy to some desired point by the application of energy at some other convenient point.
- A machine is not able to move itself and must get the motive power from some source.

Equivalent Mechanisms:

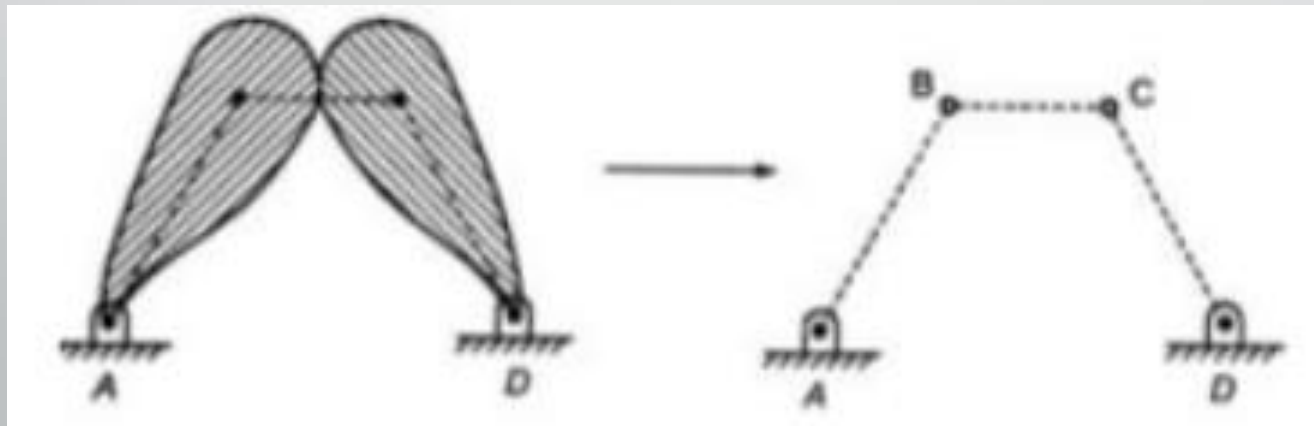
- Turning pairs of plane mechanisms may be replaced by other types of pairs such as sliding pairs or cam pairs.
 - New mechanism will be obtained having the same number of degree of freedom as the original mechanism is called the equivalent mechanism.
 - The equivalent mechanism will have same degree of freedom and shall be kinematically similar.
1. A sliding pair is equivalent to a turning pair.



2. A spring can be replaced by two binary links.



3. A cam pair can be replaced by one binary link together with two turning pairs at each end.



➤ Degree of freedom (DOF):

- ❖ An constrained rigid body moving in space can have three translations (u, v, w) and three rotational ($\theta_x, \theta_y, \theta_z$) motions about the three mutually perpendicular axes.
- ❖ Degree of freedom of a kinematic pair is defined as the number of independent relative motions, both translational and rotational.
- ❖ **Minimum number of independent coordinates (variables)** that need to be specified to fix the configuration of a mechanism.
- ❖ One link of the chain is grounded.

Degree of freedom = 6 – Number of restraints

The degree of freedom of some of systems are:

- Rigid body has six degree of freedom
- Rectangular bar sliding in a rectangular hole has one degree of freedom
- The Piston of the crank of a slider crank mechanism can be expressed by angular motion (one degree of freedom)
- A circular shaft rotating in a hole and also transmitting parallel to its axis has two degree of freedom
- A Ball and a socket joint has three degree of freedom

- Number of links = n_L
- Number of joints = n_J
- Degree of freedom of i^{th} joint = f_i

Degree of freedom of **planar mechanisms**:

$$F = 3(n_L - 1) - 3n_J + \sum_{i=1}^{n_J} f_i$$

Degree of freedom of **Spatial mechanisms**:

$$F = 6(n_L - 1) - 6n_J + \sum_{i=1}^{n_J} f_i$$

Degree of freedom (DOF)

- ❖ A mechanism must have at least 1 degree of freedom.
- ❖ A structure has zero degree of freedom.
- ❖ An over-constrained structure has negative degree of freedom.

Kutzbach criterion: The Kutbach criterion for determining the number of degree of freedom of a planar mechanism is:

$$F = 3(n - 1) - 2p - h$$

Here, F = degree of freedom

n = total number of links in a mechanism out of which one is fixed link

$n - 1$ = number of movable links

p = number of simple joints or lower pairs having one degree of freedom.

h = number of higher pairs having two degree of freedom or three degree of freedom



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$n - 1$ = number of movable links

p = number of simple joints or lower pairs having one degree of freedom.

h = number of higher pairs having two degree of freedom or three degree of freedom

Gruebler's criterion: The Gruebler's criterion for determining the number of degree of freedom of planar mechanism with lower pairs only:

$$F = 3(n - 1) - 2p$$

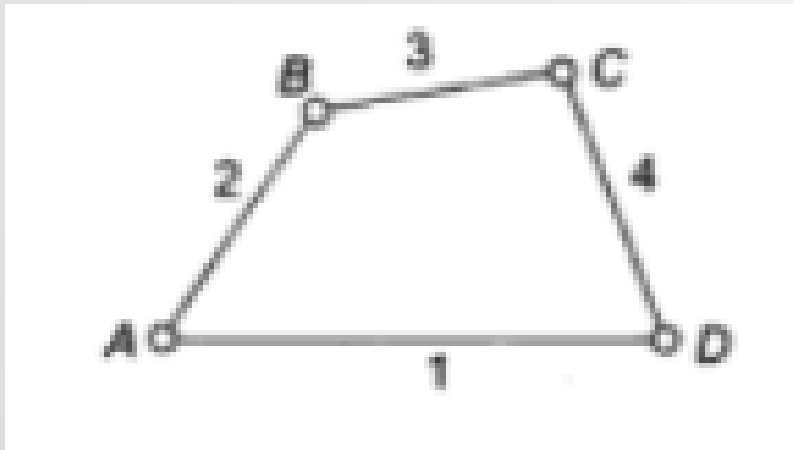
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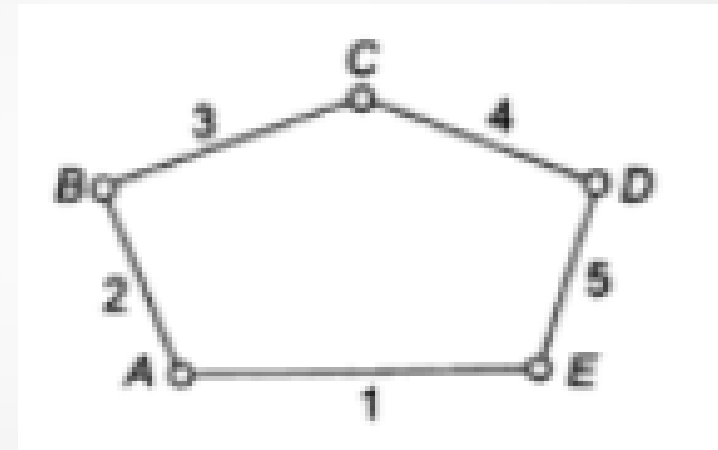
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Example 1. Calculate the number of degree of freedom of the linkage shown in Fig.



(a)



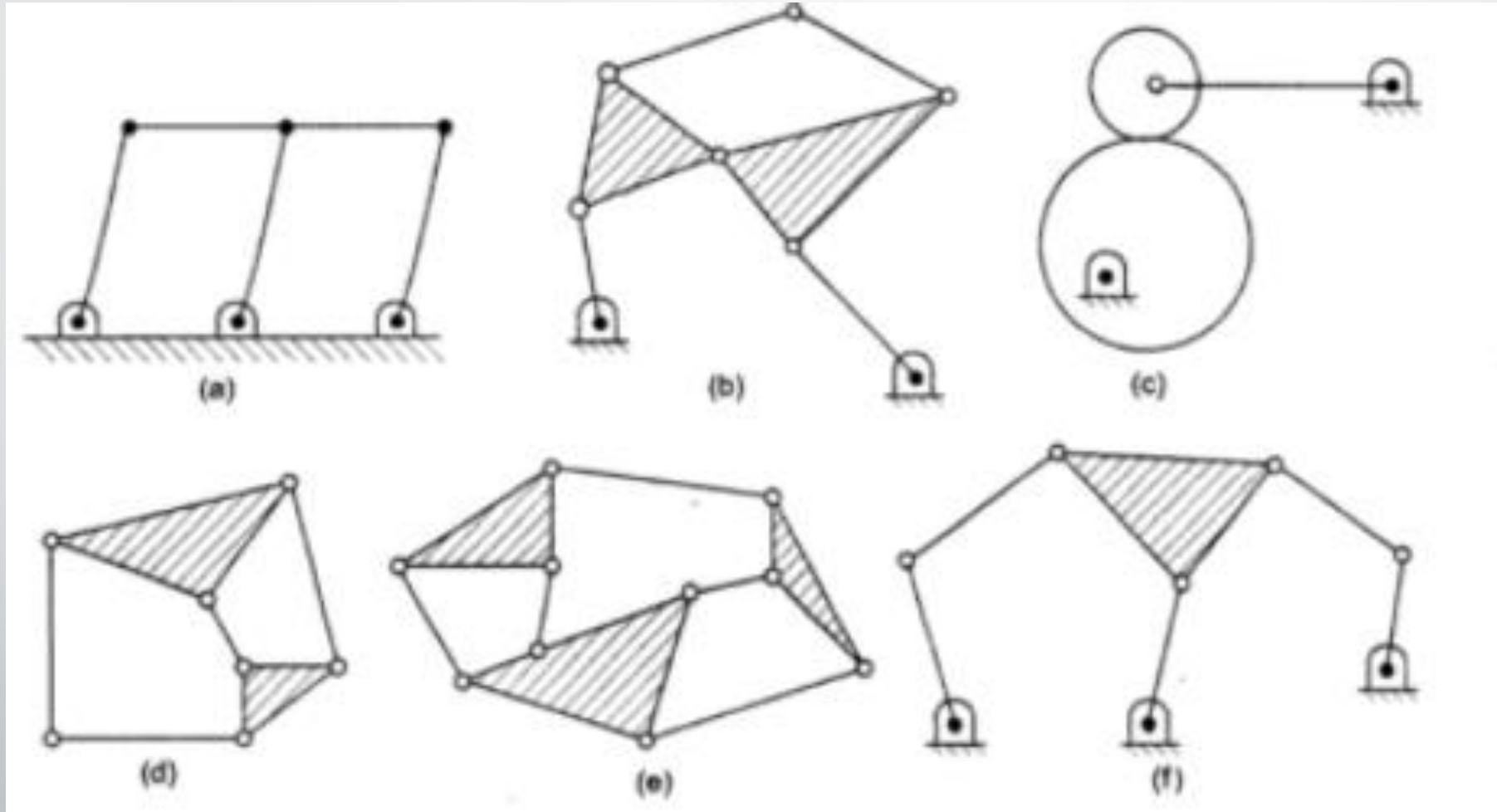
(b)

Solution:

(a) Number of binary links $n = 4$
Number of lower pairs $p = 4$
Degree of freedom $F = 3(n-1) - 2p$
 $= 3(4-1) - 2 \times 4$
 $= 1$

(b) Number of binary links $n = 5$
Number of lower pairs $p = 5$
Degree of freedom $F = 3(n-1) - 2p$
 $= 3(5-1) - 2 \times 5$
 $= 2$

Example 2. Calculate the number of degree of freedom of the linkage shown in Fig.



Solution:

(a) Number of binary links $n_2 = 3$
Number of ternary links $n_3 = 2$
Total number of links $n = n_2 + n_3 = 5$
Number of lower pairs $2p = 2n_2 + 3n_3$
 $= 2 \times 3 + 3 \times 2$
 $= 12$
Degree of freedom $F = 3(n-1) - 2p$
 $= 3(5-1) - 12$
 $= 0$

(b) Number of binary links $n_2 = 5$
Number of ternary links $n_3 = 2$
Total number of links $n = n_2 + n_3 = 7$
Number of lower pairs $2p = 2n_2 + 3n_3$
 $= 2 \times 5 + 3 \times 2$
 $= 16$
Degree of freedom $F = 3(n-1) - 2p$
 $= 3(7-1) - 16$
 $= 2$

Solution:

(c) Number of binary links $n_2 = 3$

Number of ternary links $n_3 = 0$

Total number of links $n = n_2 + n_3 = 5$

Number of lower pairs $p = 2$

Number of higher pairs $h = 1$

Degree of freedom $F = 3(n-1) - 2p - h$
 $= 3(5-1) - 4 - 1$
 $= 1$

(d) Number of binary links $n_2 = 4$

Number of ternary links $n_3 = 2$

Total number of links $n = n_2 + n_3 = 6$

Number of lower pairs $2p = 2n_2 + 3n_3$
 $= 2 \times 4 + 3 \times 2$
 $= 14$

Degree of freedom $F = 3(n-1) - 2p$
 $= 3(6-1) - 14$
 $= 1$

Solution:

(e) Number of binary links $n_2 = 5$

Number of ternary links $n_3 = 2$

Number of quaternary links $n_4 = 1$

Total number of links $n = n_2 + n_3 + n_4 = 8$

Number of lower pairs $2p = 2n_2 + 3n_3 + 4n_4$
 $= 2 \times 5 + 3 \times 2 + 4 \times 1$
 $= 20$

Degree of freedom $F = 3(n-1) - 2p$
 $= 3(8-1) - 20$
 $= 1$

(f) Number of binary links $n_2 = 5$

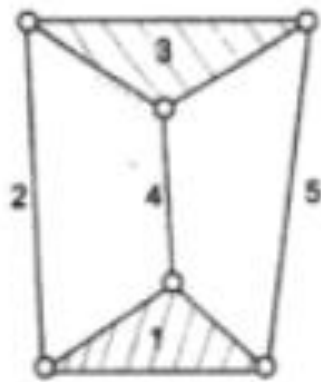
Number of ternary links $n_3 = 2$

Total number of links $n = n_2 + n_3 = 7$

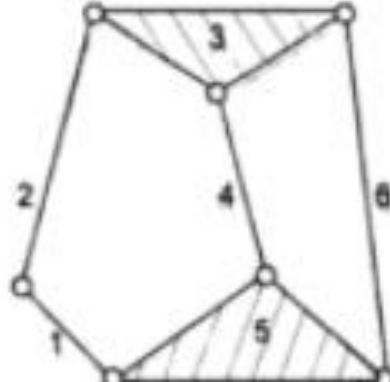
Number of lower pairs $2p = 2n_2 + 3n_3$
 $= 2 \times 5 + 3 \times 2$
 $= 16$

Degree of freedom $F = 3(n-1) - 2p$
 $= 3(7-1) - 16$
 $= 2$

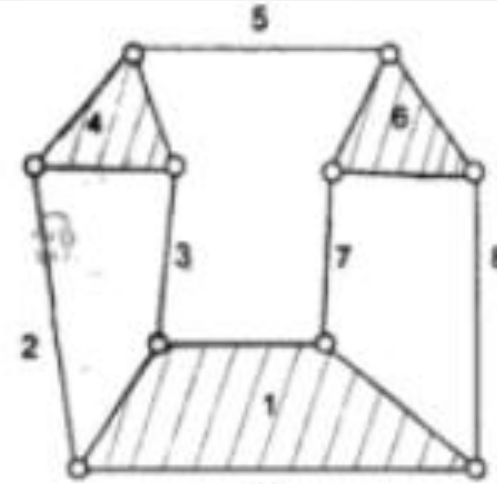
Example 3. Calculate the number of degree of freedom of the linkage shown in Fig. (for practice).



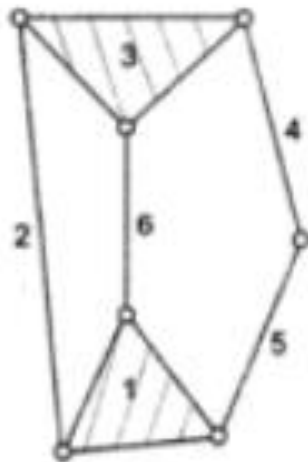
(a)



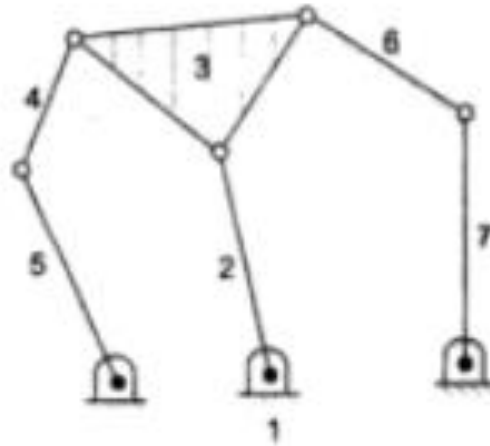
(b)



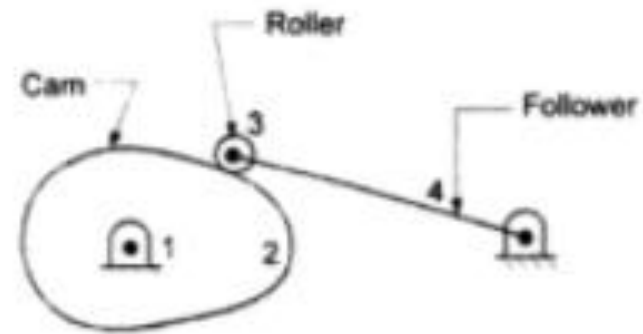
(c)



(d)



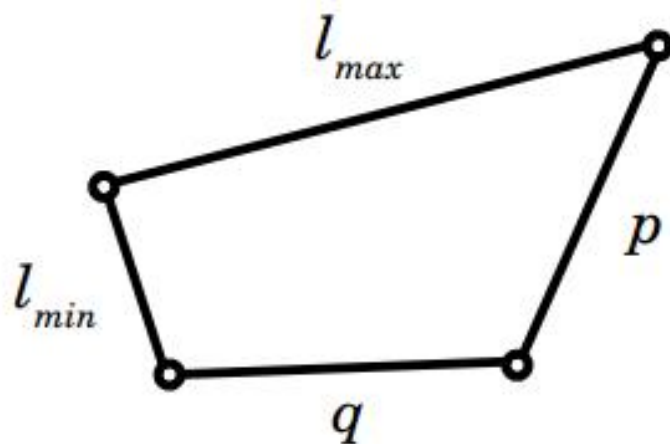
(e)



(f)

Grashof's Law or Grashof criterion (4R Chain):

This law states that for a four-bar mechanism the sum of the length of the largest and the shortest links should be less than or equal to the sum of the lengths of the other links.



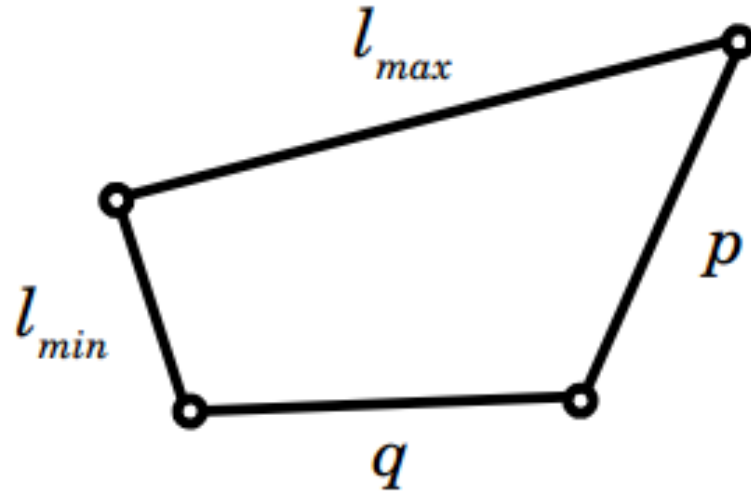
- l_{min} : length of shortest link
- l_{max} : length of longest link
- p and q : length of other two links

If

$$l_{min} + l_{max} \leq p + q$$

then the shortest link can rotate completely with respect to all other links

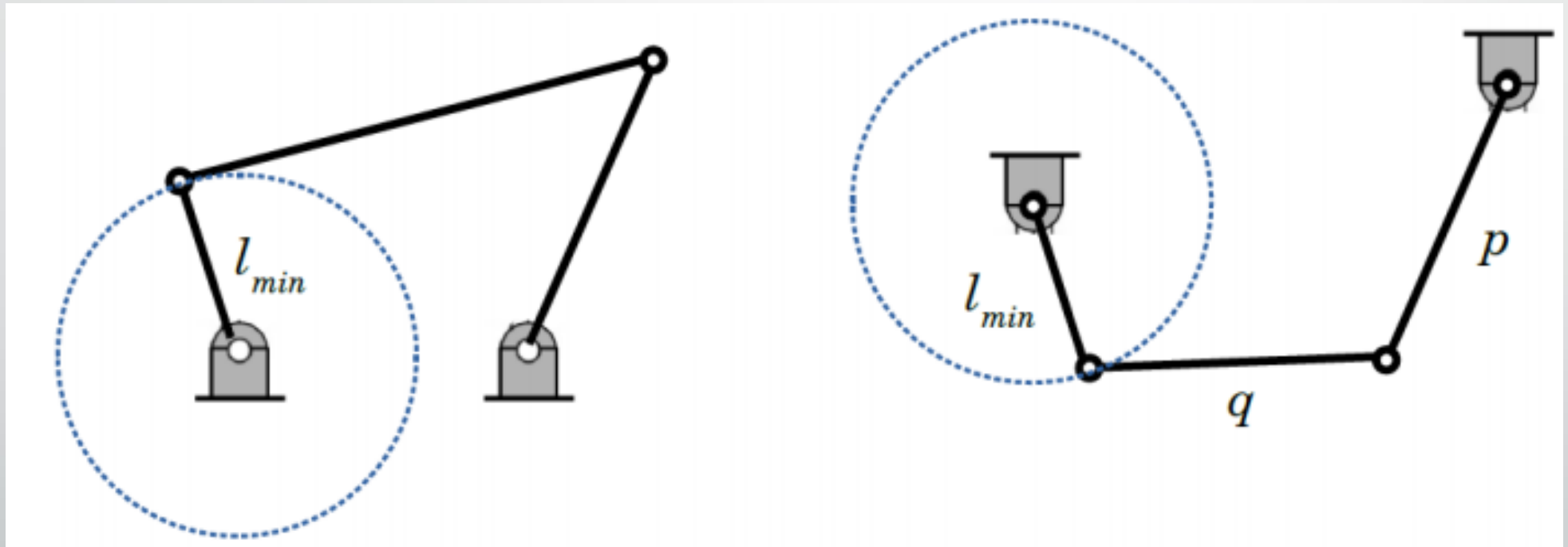
Grashof 4R chain



- Shortest link rotates with respect to all other links
- All other links rock with respect to each other

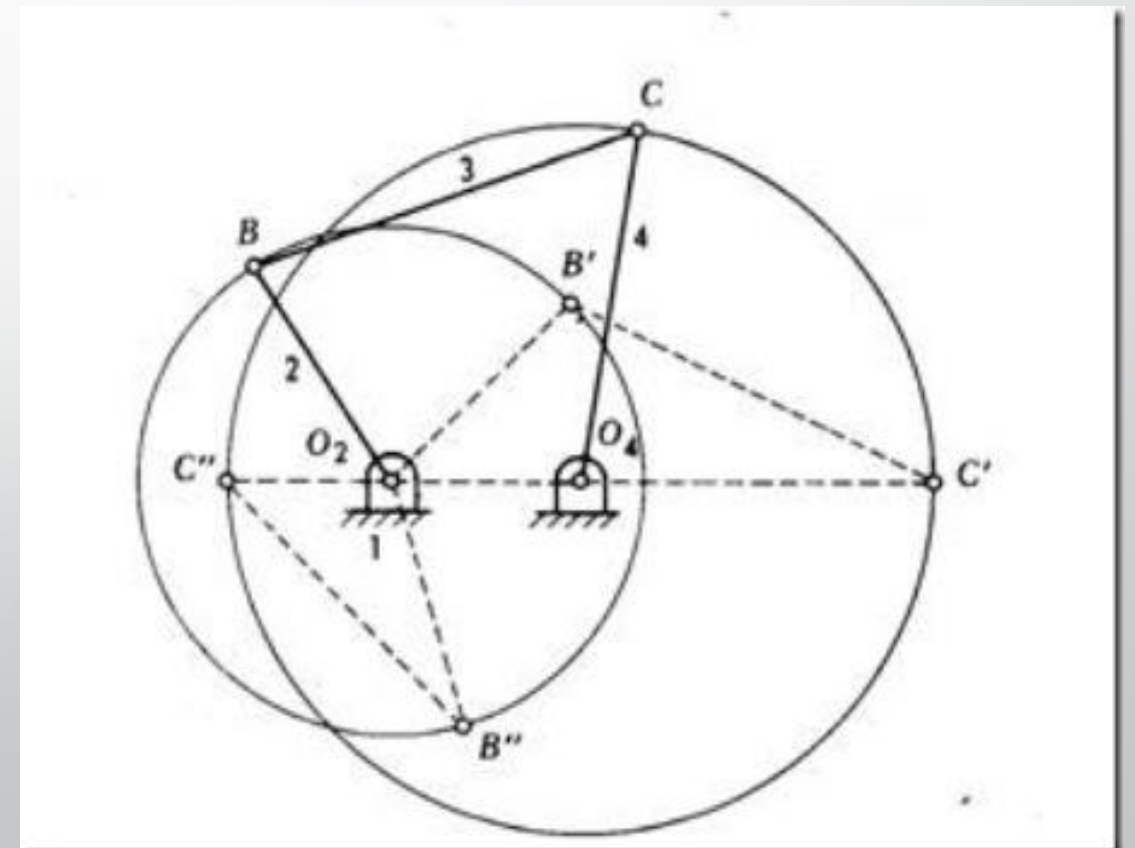
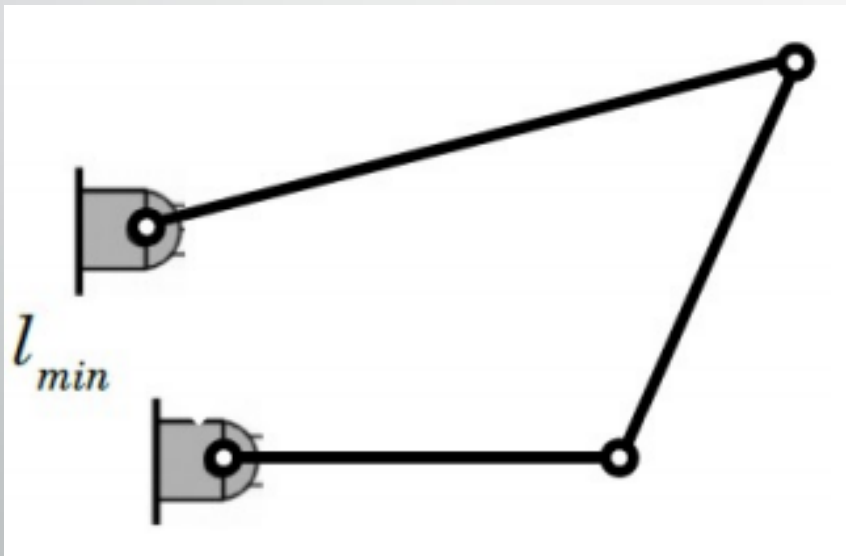
Inversions of for-bar chain: various possibilities depending on choice of ground link.

1. Crank-rocker mechanism



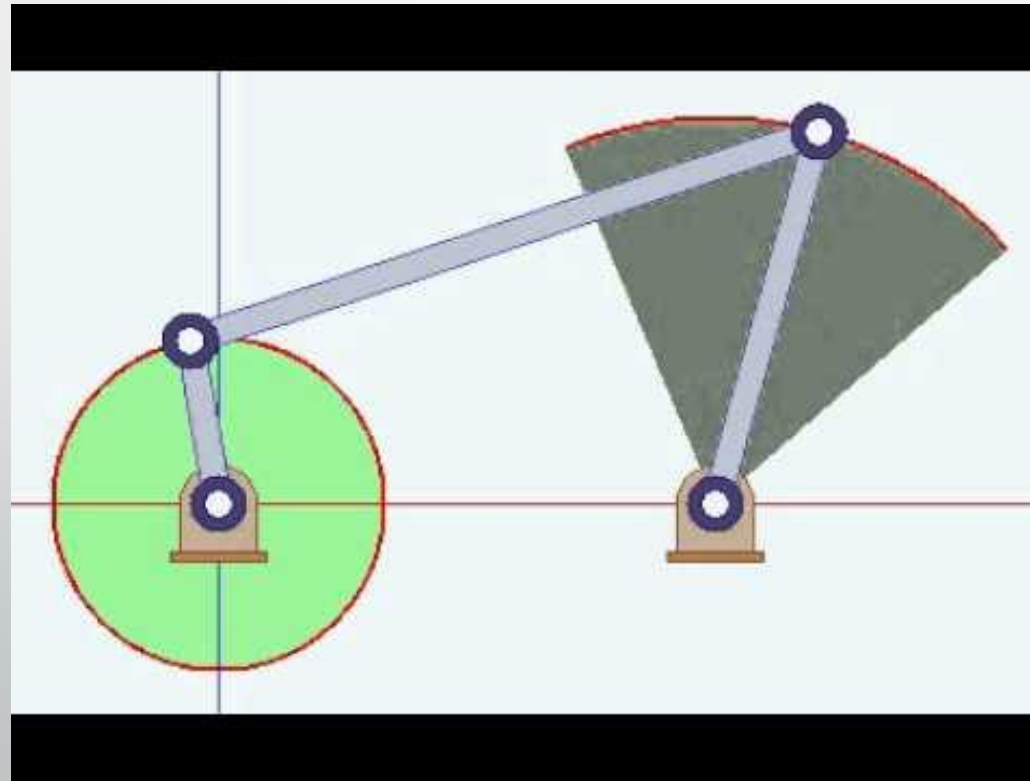
Inversions of for-bar chain: various possibilities depending on choice of ground link.

2. Crank-Crank (or double crank) mechanism: when the shortest link is fixed and $l_{min} + l_{max} \leq p + q$, crank-crank or double crank mechanism is obtained.



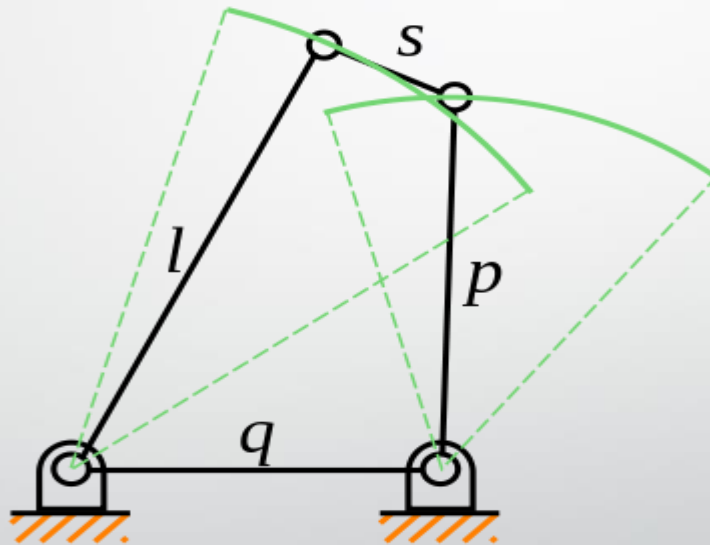
Inversions of for-bar chain: various possibilities depending on choice of ground link.

3. Crank-Rocker (or lever) mechanism: if any of the adjacent link of shortest link is fixed than shortest link have full revolution and the link L3 opposite to shortest link can oscillates.

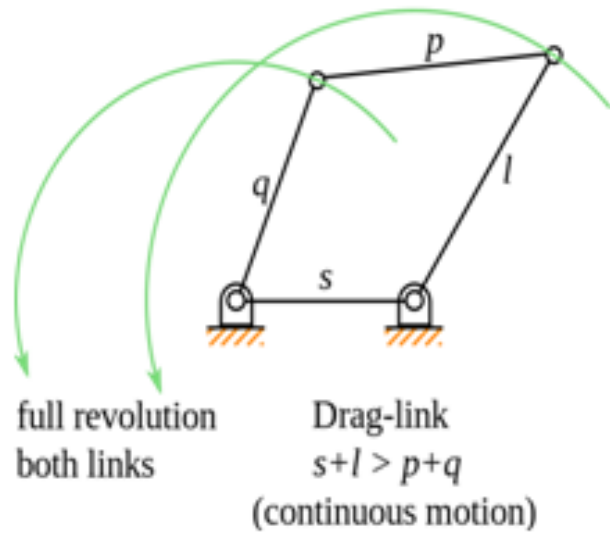


Inversions of for-bar chain: various possibilities depending on choice of ground link.

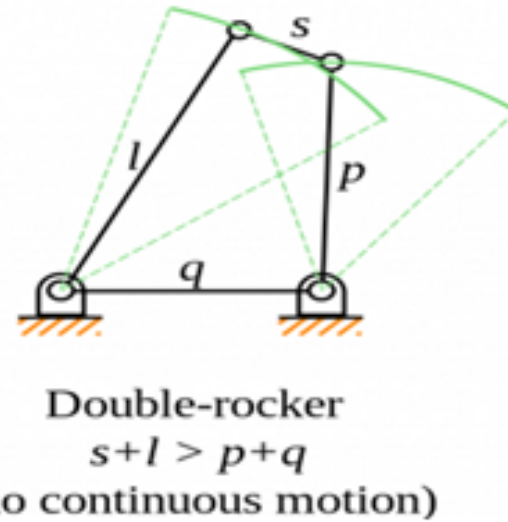
4. Rocker-Rocker (or double rocker) mechanism: if link L3 opposite to shortest link L1 is fixed and shortest link L1 is made the coupler, the other two links L2 and L4 would oscillate.



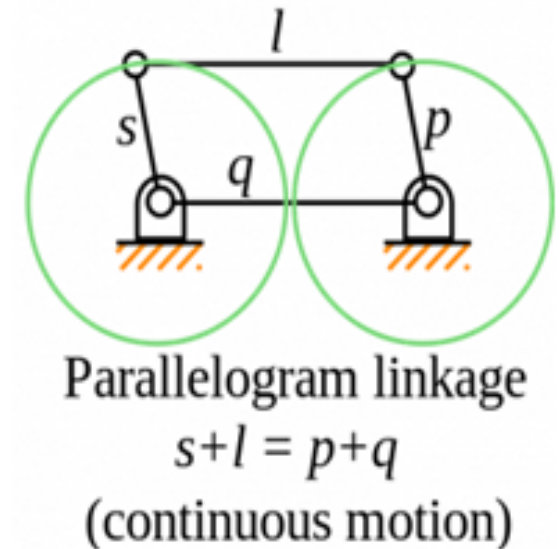
Double-rocker
 $s+l > p+q$
(no continuous motion)



(a) Double crank mechanism



(b) Double rocker mechanism



(c) Parallelogram linkage

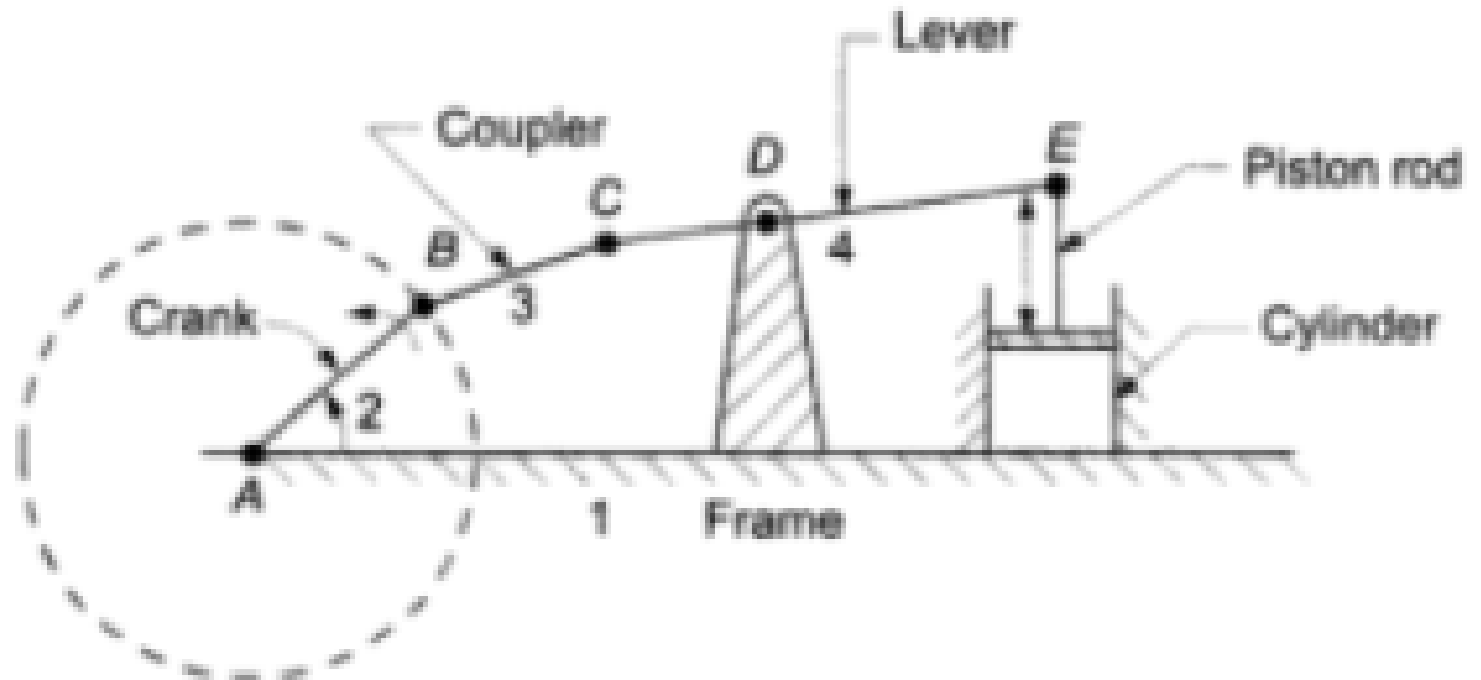
Fig. 4 Possible mechanism according to Grashof's law

Inversions of Mechanisms:

- A kinematic chain becomes a mechanism when one of the link is fixed.
- Therefore, as many number of mechanisms can be obtained as many are the links in the kinematic chain.
- This method of obtaining different mechanisms by fixing different links of a kinematic chain is called inversion of the mechanism.
- The relative motion between the various links is not altered as a result of inversion, but their absolute motion with respect to the fixed link may alter drastically.

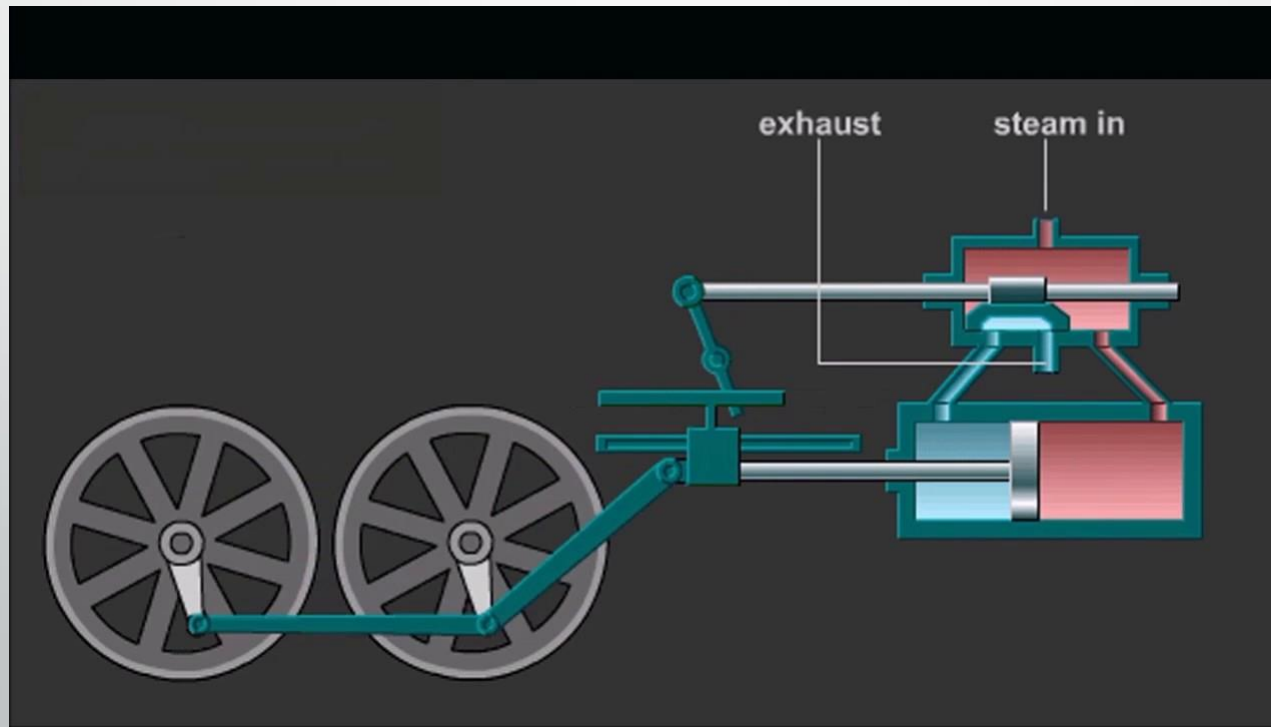
Inversions of a for-bar chain are:

- Beam engine (crank and lever mechanism): a beam engine is a type of steam engine where a pivoted overhead beam is used to apply the force from a vertical piston to a vertical connecting rod.
- Beam engines were first used to pump water out of mines or into canals.



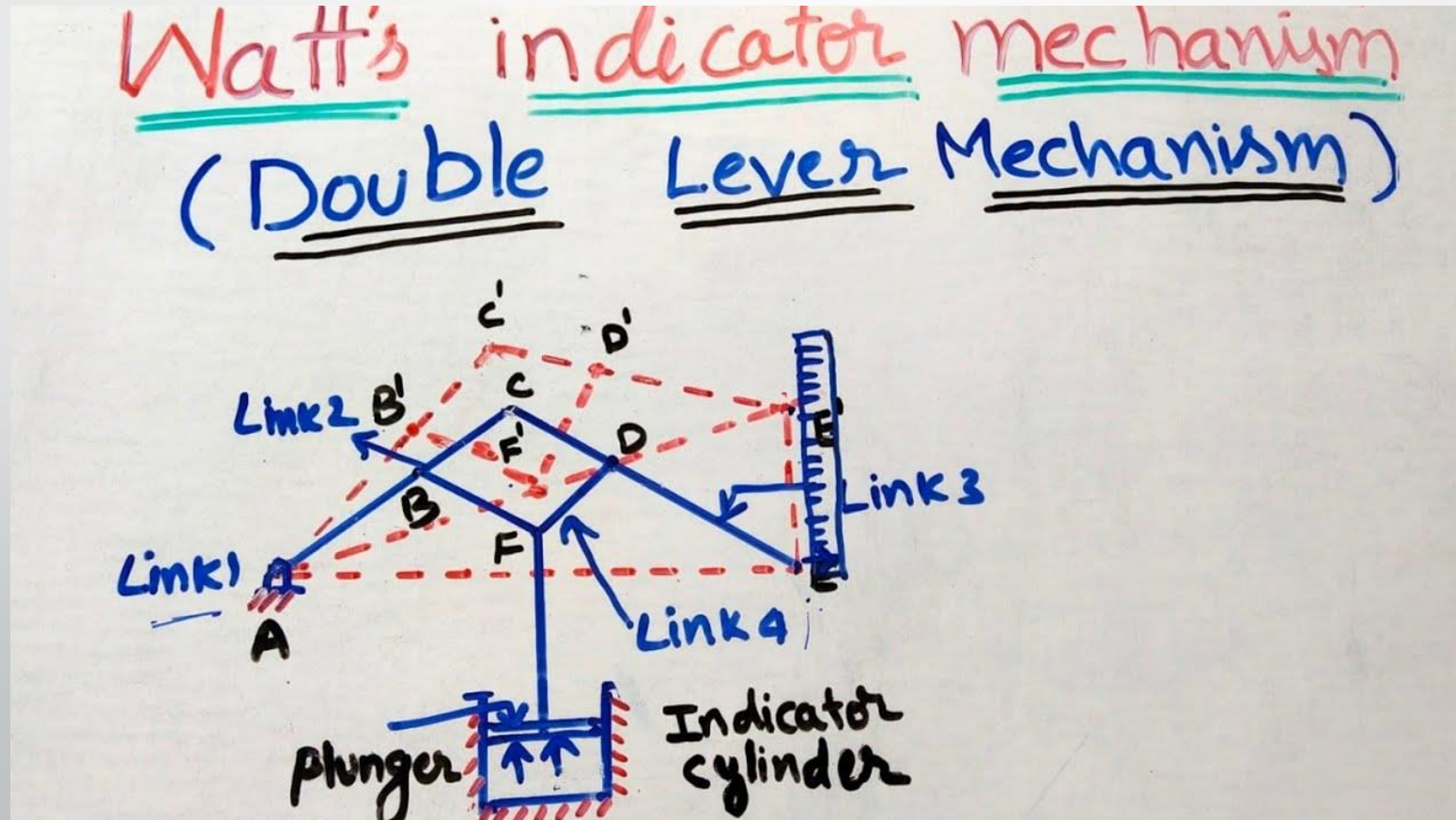
Coupled wheel of locomotive:

- The links AB and CD are of equal lengths and act as crank.
- These cranks are connected to the respective wheels.
- This mechanism is used to transmit rotary motion from one wheel to the other.
- This is also called the **double crank mechanism**.



Inversions of a for-bar chain are:

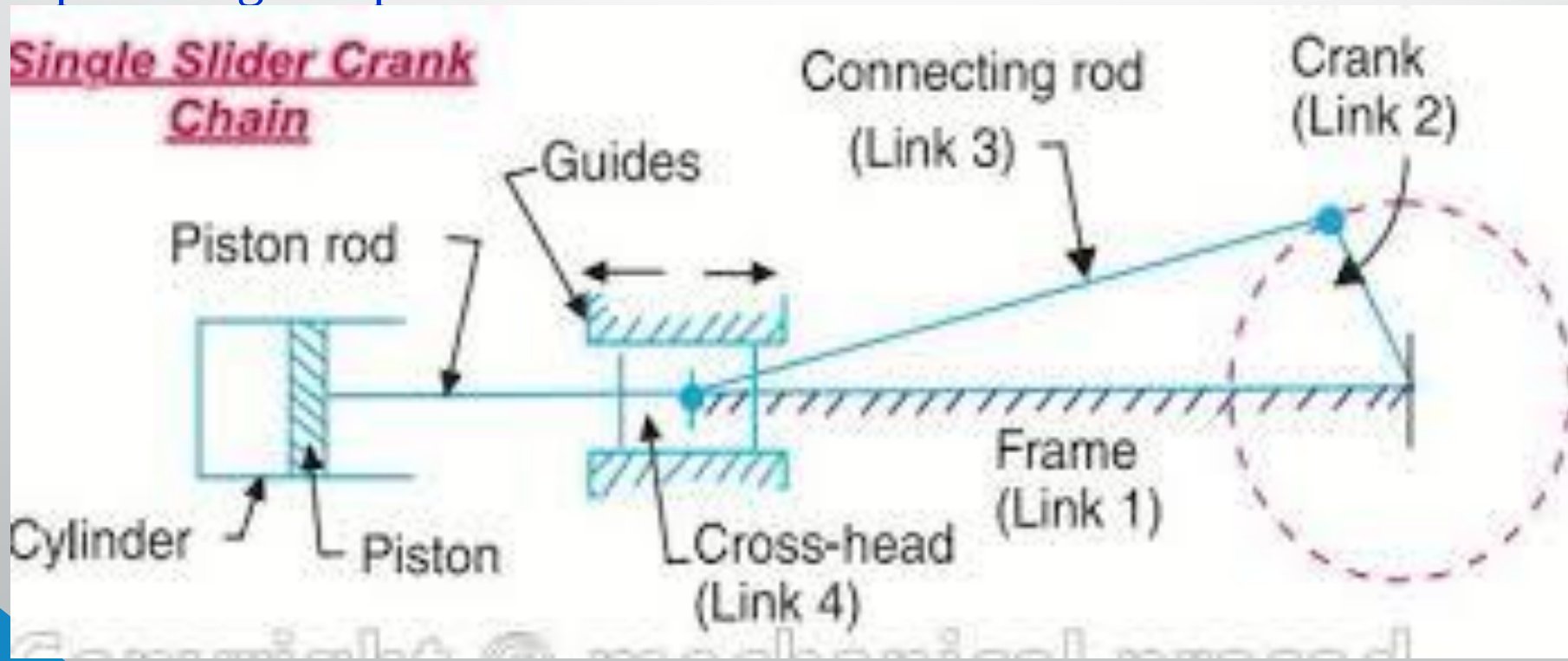
- Watt's indicator mechanism: it consists of four links: a fixed link at A, link AC, link CE, and link BFD.
- The links CE and BFD acts as levers.
- The displacement of link BFD is directly proportional to the pressure in the indicator cylinder.



Inversions of a for-bar chain are:

Single slider-crank chain: it consists of three turning pairs and one sliding pair.

- It is used to convert rotary motion to reciprocating and vice-versa.
- Applications: steam engine, internal combustion engine, reciprocating compressor.





Thank You

ME-203A: Kinematics of Machines

by

Dr. Emarti Kumari



**Department of Mechanical Engineering
M.B.M. Engineering College
Jai Narain Vyas University, Jodhpur**

UNIT 1

Kinematic analysis of mechanism: Definitions of Links, pairs and kinematic chain, Mechanism and machine, Inversion of four bar kinematic chain and slider crank chains.

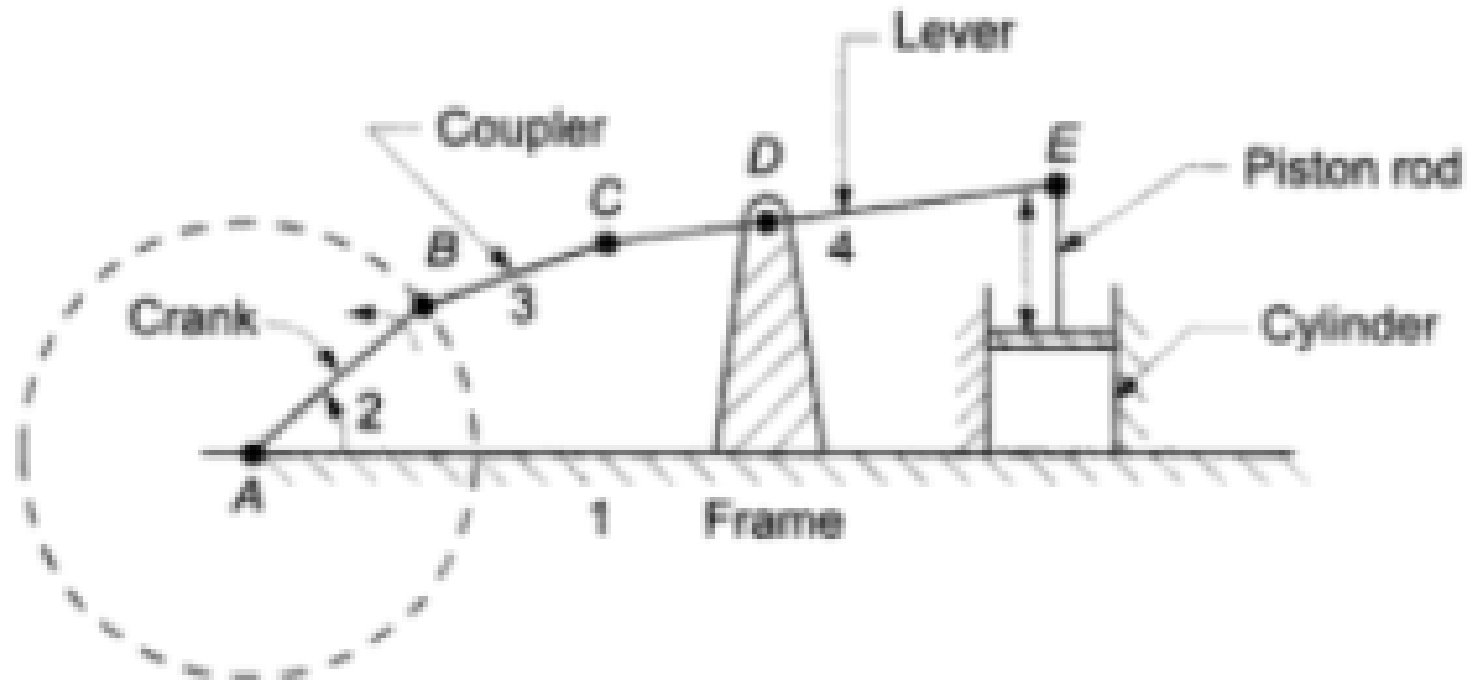
Velocity and acceleration analysis: Relative velocity method; Instantaneous center method, Coriolis component of acceleration

Inversions of Mechanisms:

- A kinematic chain becomes a mechanism when one of the link is fixed.
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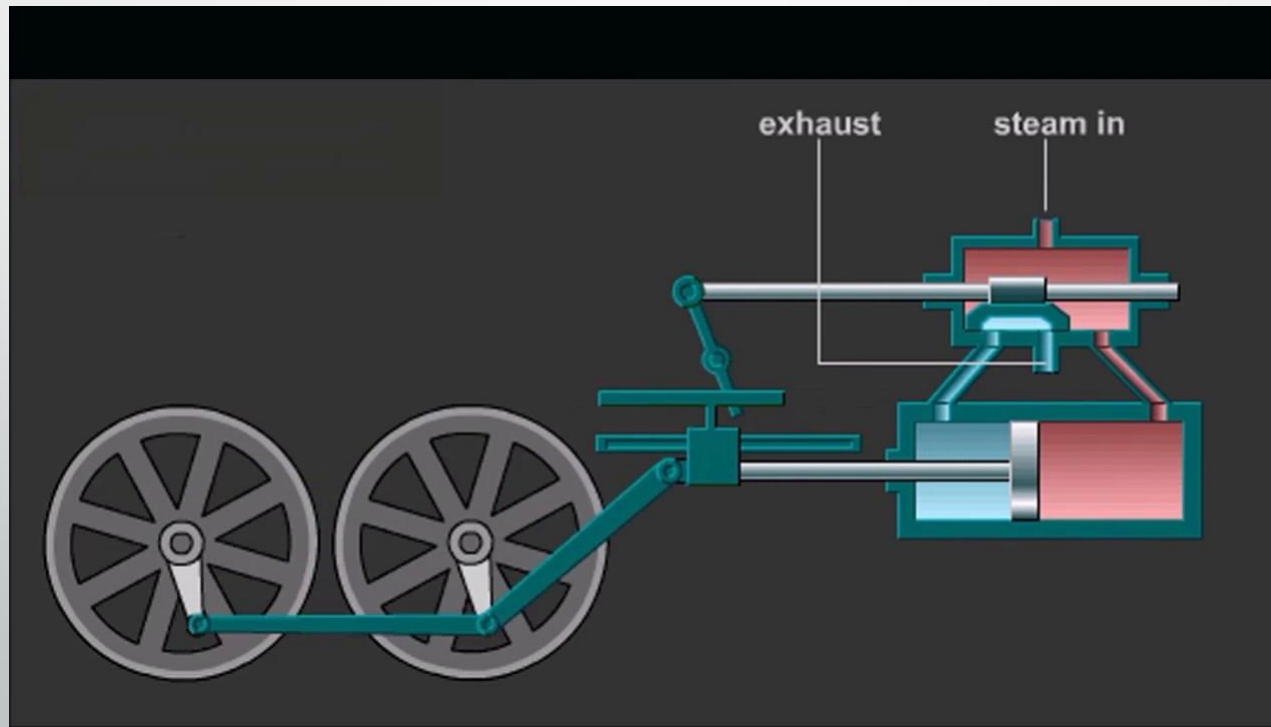
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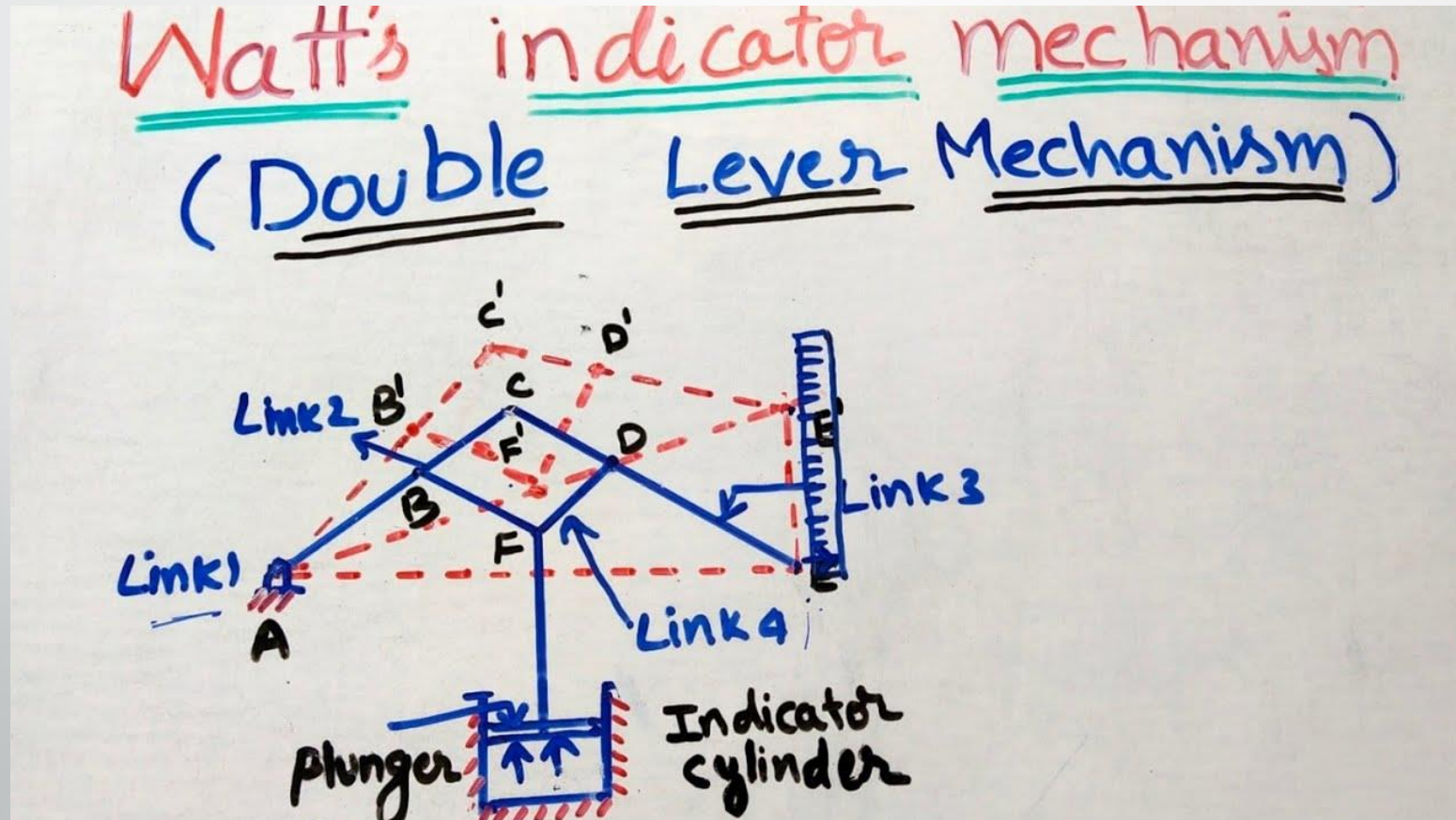
Coupled wheel of locomotive:

- The links AB and CD are of equal lengths and act as crank.
- These cranks are connected to the respective wheels.
- This mechanism is used to transmit rotary motion from one wheel to the other.
- This is also called the double **crank mechanism**.



Inversions of a for-bar chain are:

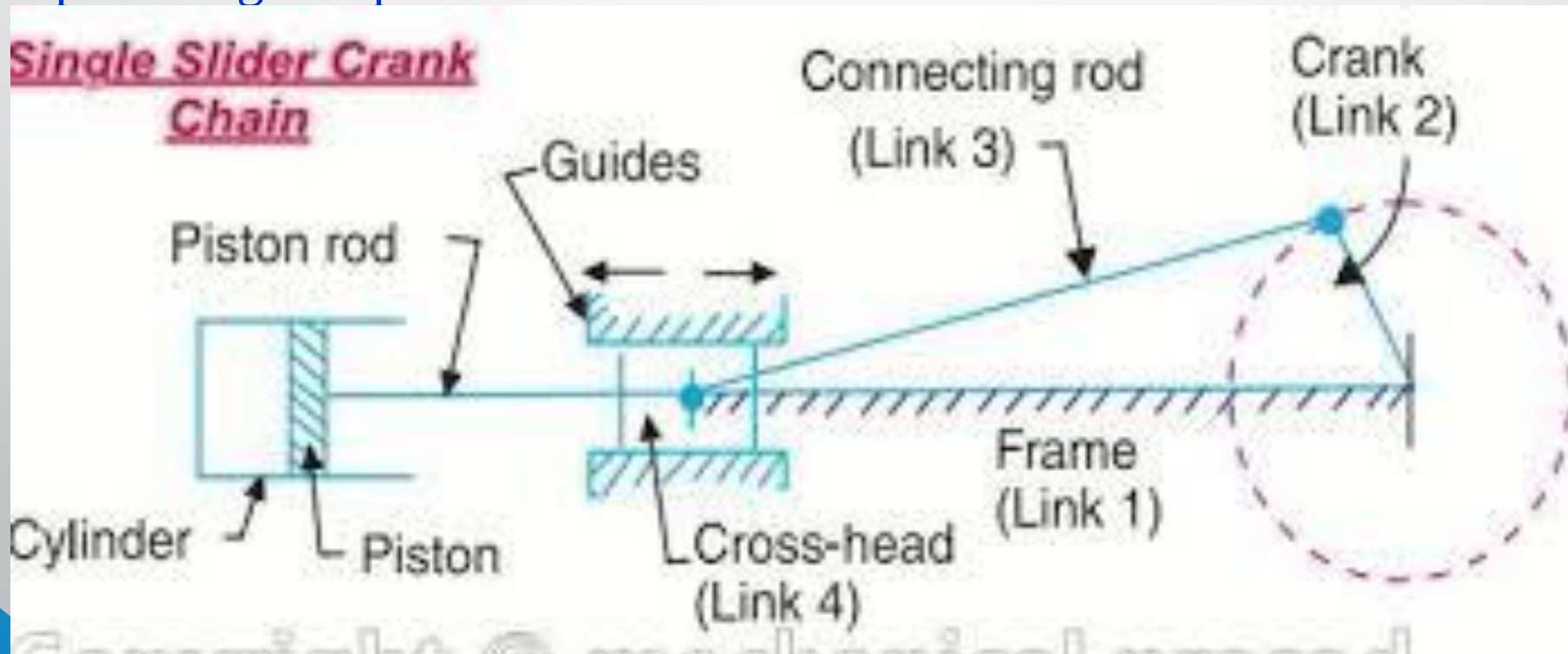
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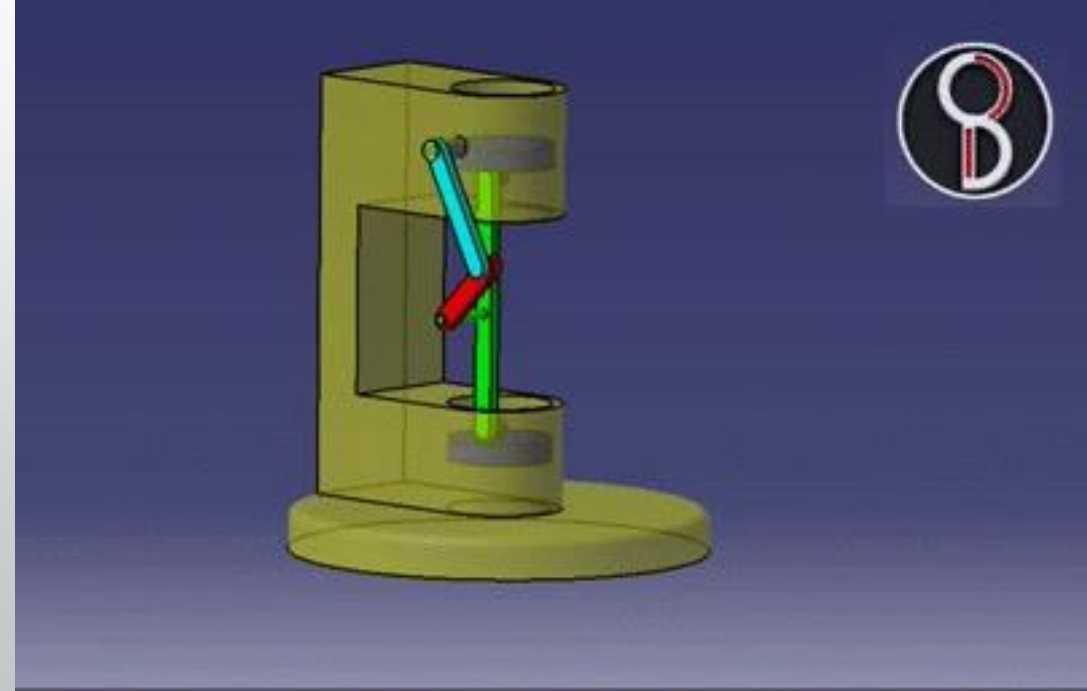
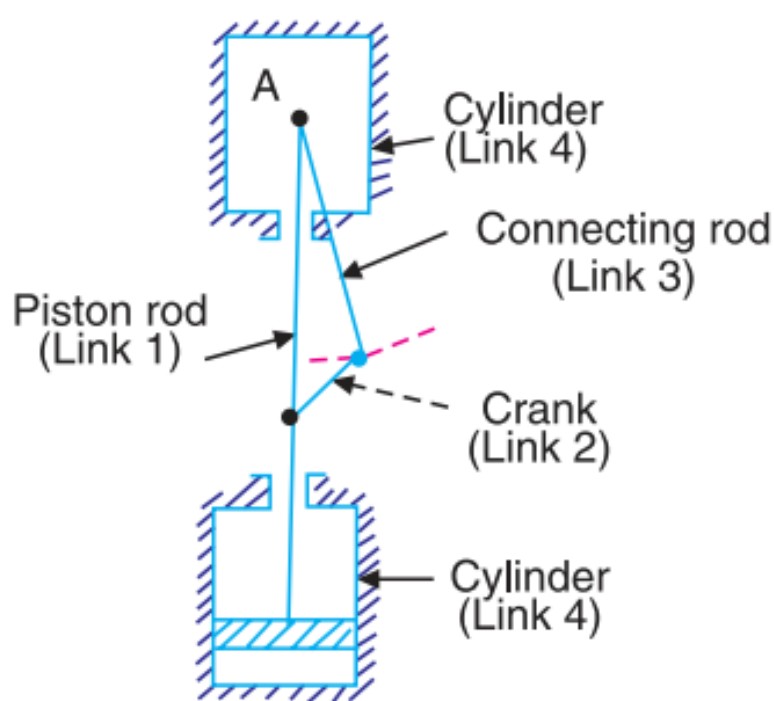
Single slider-crank chain: it consists of three turning pairs and one sliding pair.

- It is used to convert rotary motion to reciprocating and vice-versa.
- Applications: steam engine, internal combustion engine, reciprocating compressor.



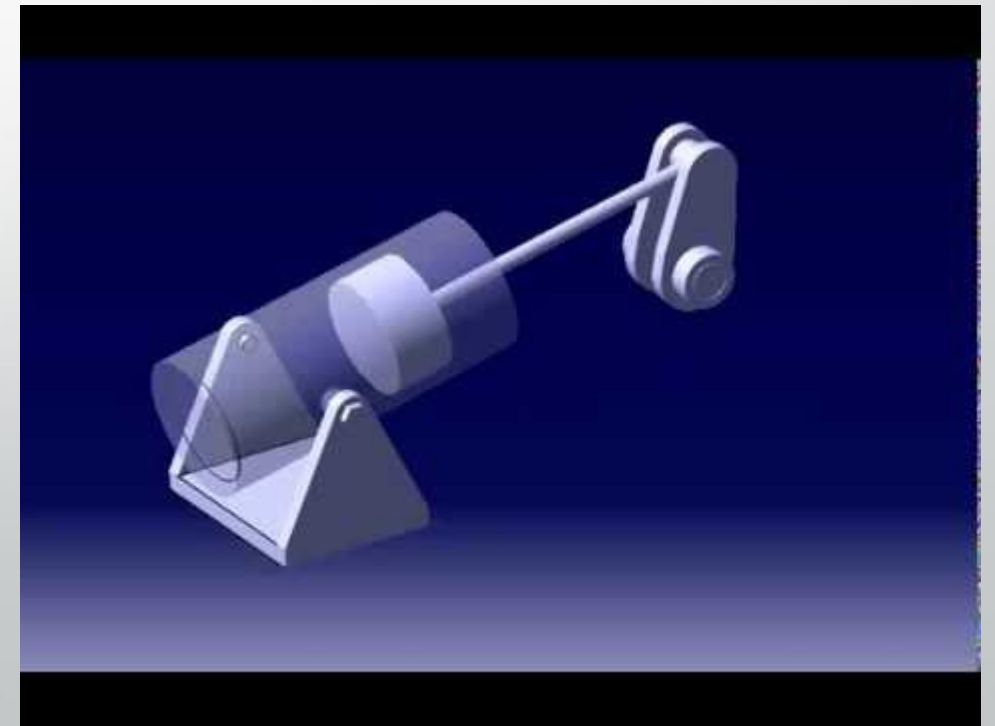
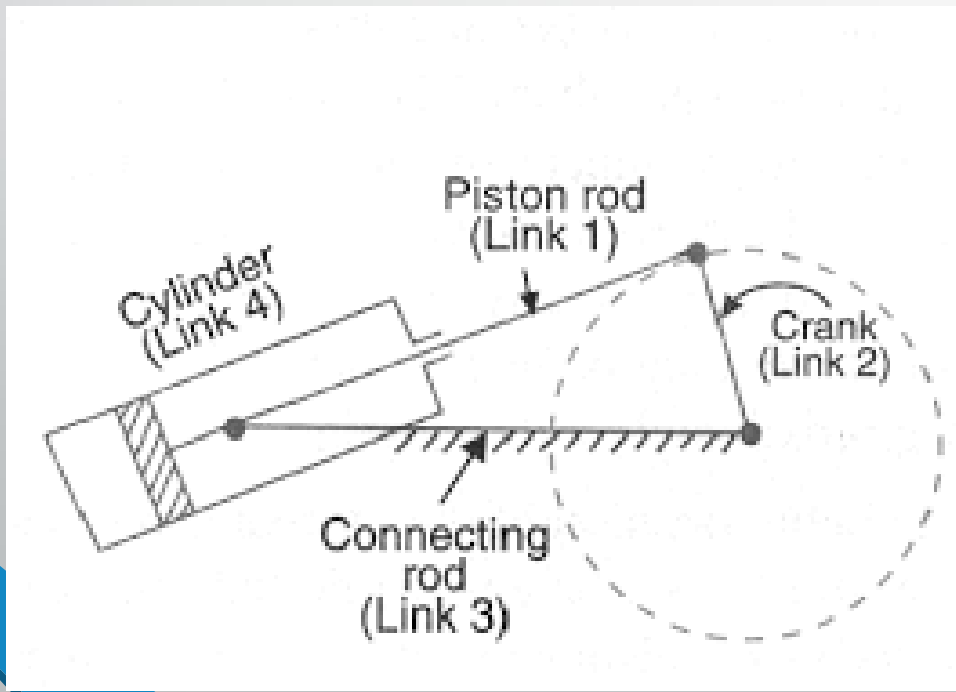
Inversions of a Single-Slider Crank chain are:

- **Pendulum Pump:** this inversion mechanism is obtained by fixing the link 4.
- When link 2 (crank) rotates, the link 3 (connecting rod) oscillates about the a pin pivoted to fixed link 4 at A and the piston attached to the piston rod (link 1) reciprocates in the cylinder.



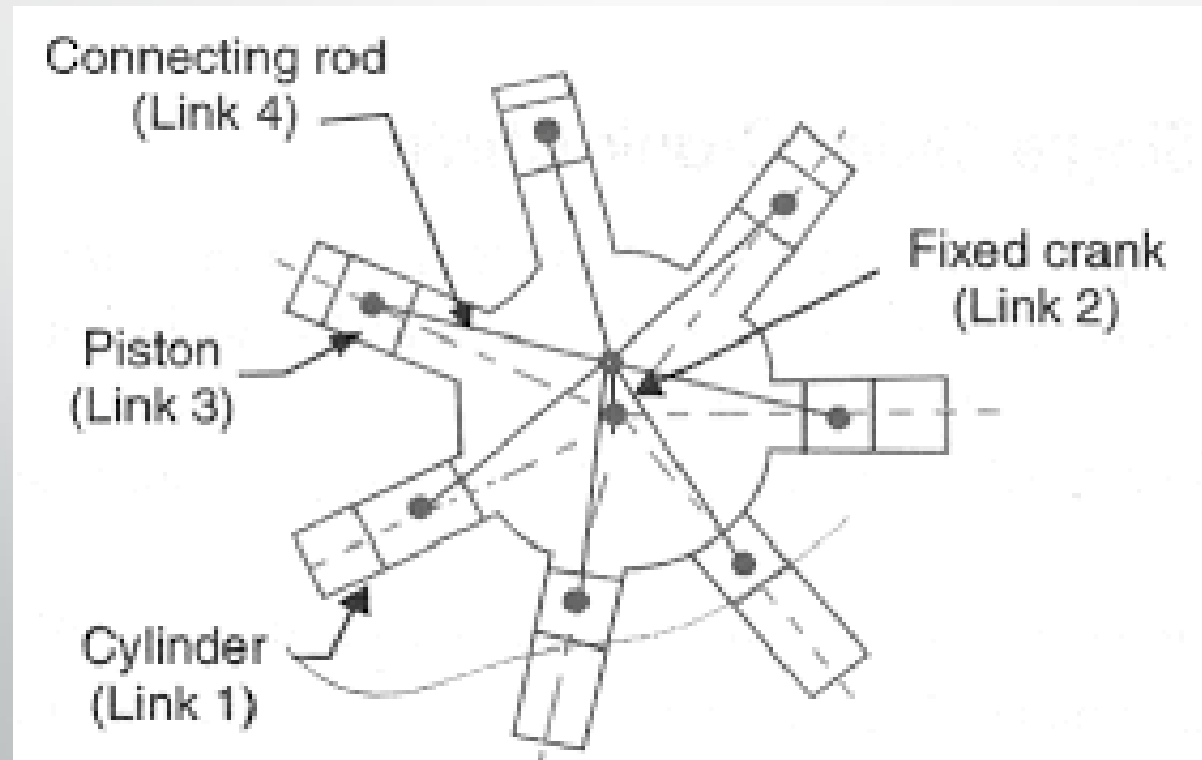
Inversions of a Single-Slider Crank chain are:

- **Oscillating cylinder engine:** In this mechanism link 3 is fixed.
- When the crank (link 2) rotates, the piston attached to piston rod (link 1) reciprocates and the cylinder (link 4) oscillates about a pin pivoted to the fixed link at A.



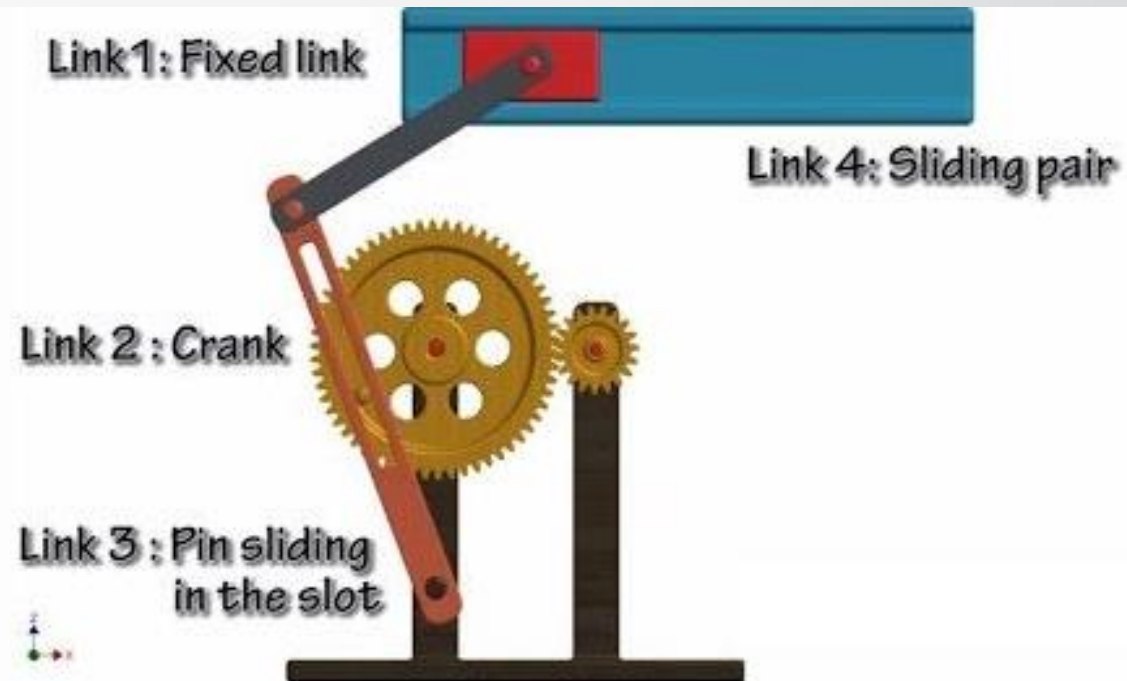
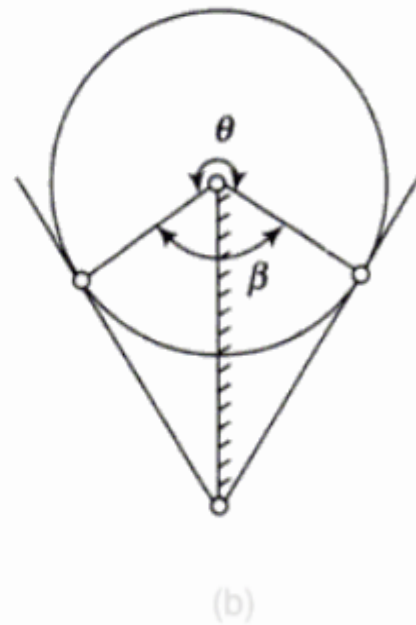
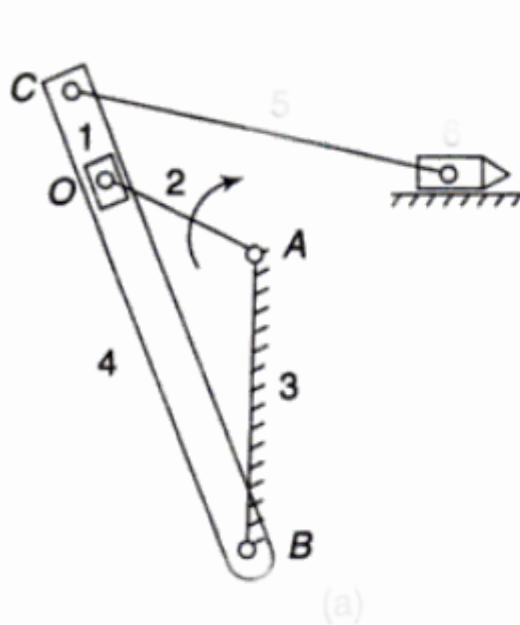
Inversions of a Single-Slider Crank chain are:

- **Rotary internal combustion engine (Gnome engine):**
It consists of several cylinder in one plane and all revolve about fixed centre O.
- The crank (link 2) is fixed.
- When the connecting rod (link 4) rotates, the piston (link 3) reciprocates inside the cylinder forming link 1.



Inversions of a Single-Slider Crank chain are:

- **Crank and slotted lever quick-return motion mechanism:** In this mechanism link 3 corresponding to the connecting rod is fixed.
- The driving crank OA (link 2) revolves about centre O.
- A slider (link 1) attached to the crank pin at B slides along the slotted lever CB (link 4) and make the slotted lever oscillate about the pivoted point A.



Inversions of a Single-Slider Crank chain are:

- **Crank and slotted lever quick-return motion mechanism:** In this mechanism link 3 corresponding to the connecting rod is fixed.
- The driving crank OA (link 2) revolves about centre O.
- A slider (link 1) attached to the crank pin at B slides along the slotted lever CB (link 4) and make the slotted lever oscillate about the pivoted point A.
- A short link CD transmits the motion from BC to the arm which reciprocates with the tool along the line of stroke.
- The line of stroke is perpendicular to AB produced.
- This mechanism is mostly used in shaping machines, slotting machine, and rotary internal combustion engines.

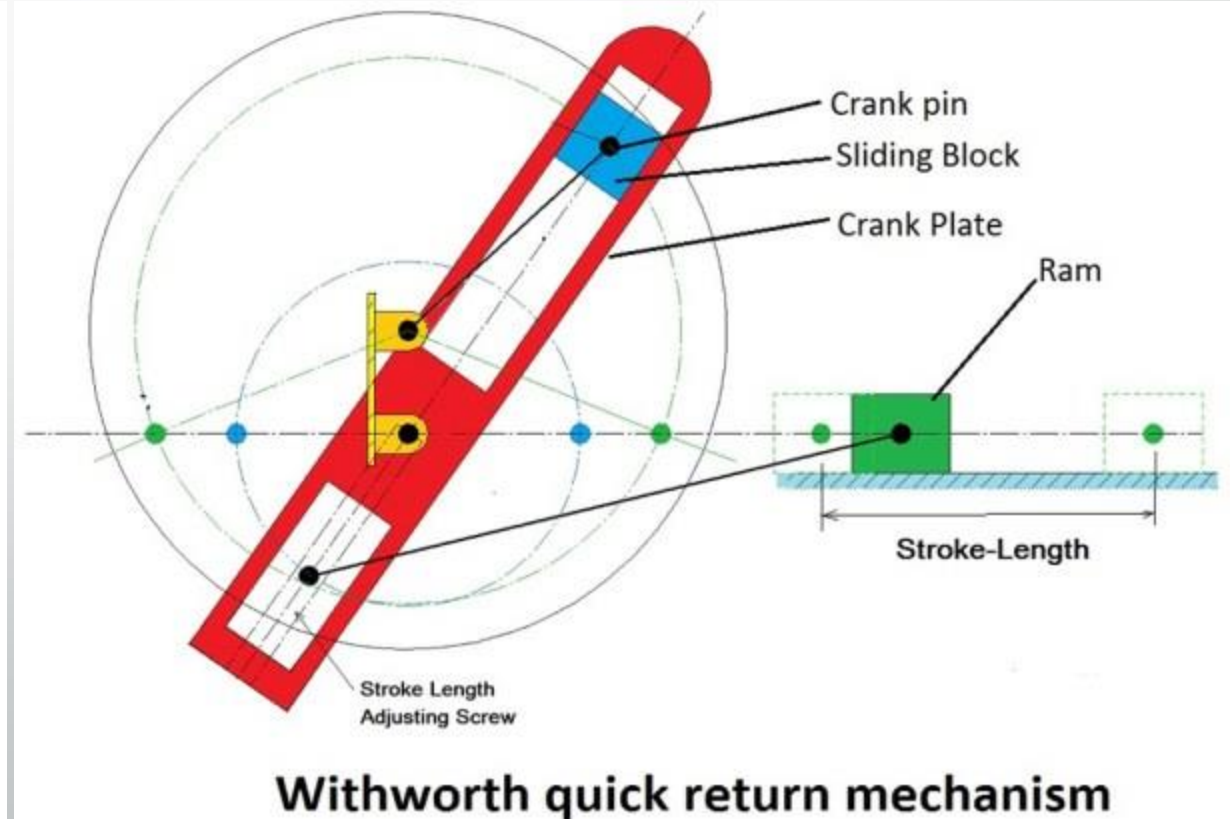
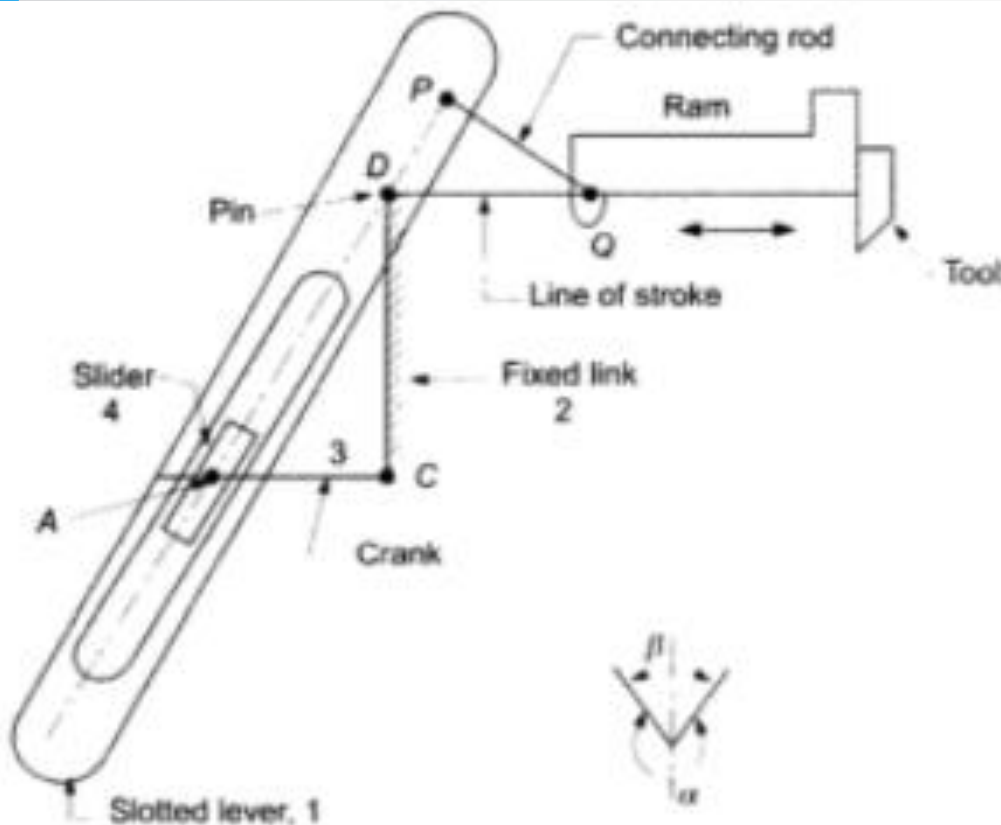


Time of cutting stroke / Time of return stroke = α / β

Length of stroke = 2 BC (AO / AB)

Inversions of a Single-Slider Crank chain are:

- **Whitworth quick-return motion mechanism:** In this mechanism link CD (link 2) is fixed.
- The driving crank CA (link 3) rotates about C.
- The slider (link 4) attached to the pin at A slides along the slotted lever PA (link 1), which oscillates about pivot D.



Inversions of a Single-Slider Crank chain are:

- **Whitworth quick-return motion mechanism:** In this mechanism link CD (link 2) is fixed.
- The driving crank CA (link 3) rotates about C.
- The slider (link 4) attached to the pin at A slides along the slotted lever PA (link 1), which oscillates about pivot D.
- The connecting rod PQ carries the ram at Q with cutting tool.
- The ram reciprocates along the line of stroke,
- It is used in shaping and slotting machines.

$$\text{Time of cutting stroke} / \text{Time of return stroke} = \alpha / \beta$$

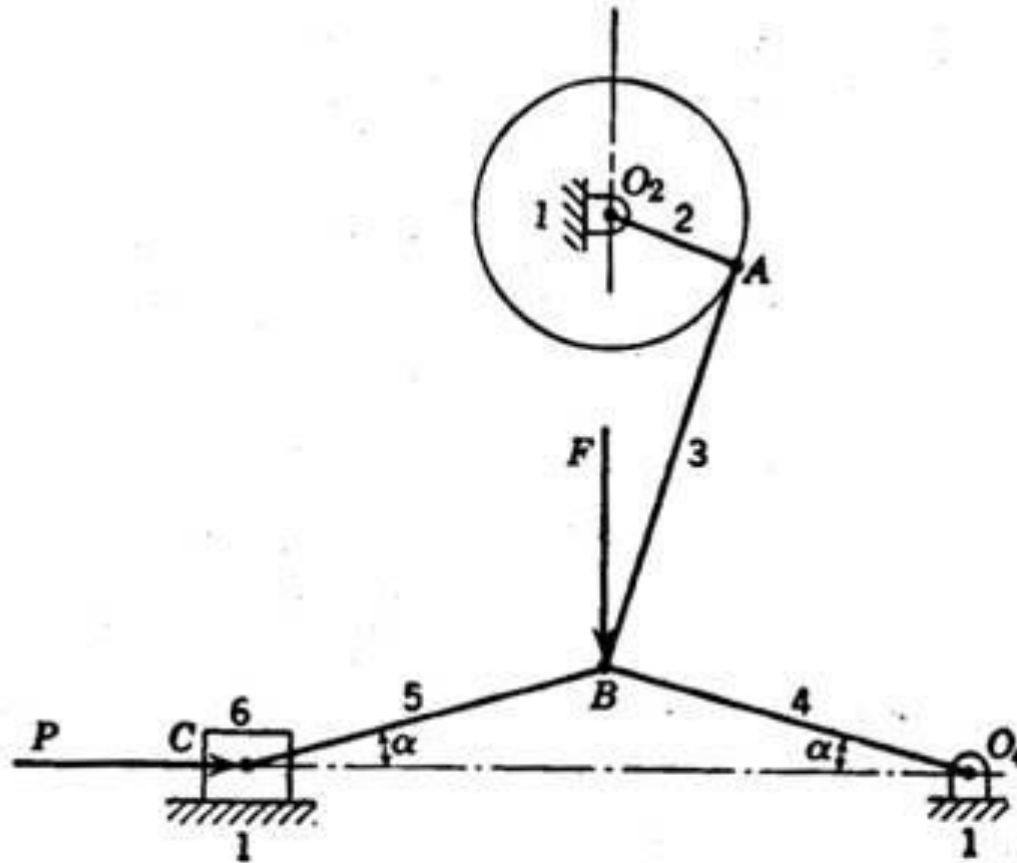
$$\text{Length of stroke} = 2 PD$$

Inversions of a Single-Slider Crank chain are:

- **Toggle mechanism:** This mechanism has many applications where it is necessary to overcome a large resistance with a small driving force.
- Links 4 and 5 are equal length.
- As the angles α decrease and links 4 and 5 approach being collinear, the force F required to overcome a given resistance P decreases as:

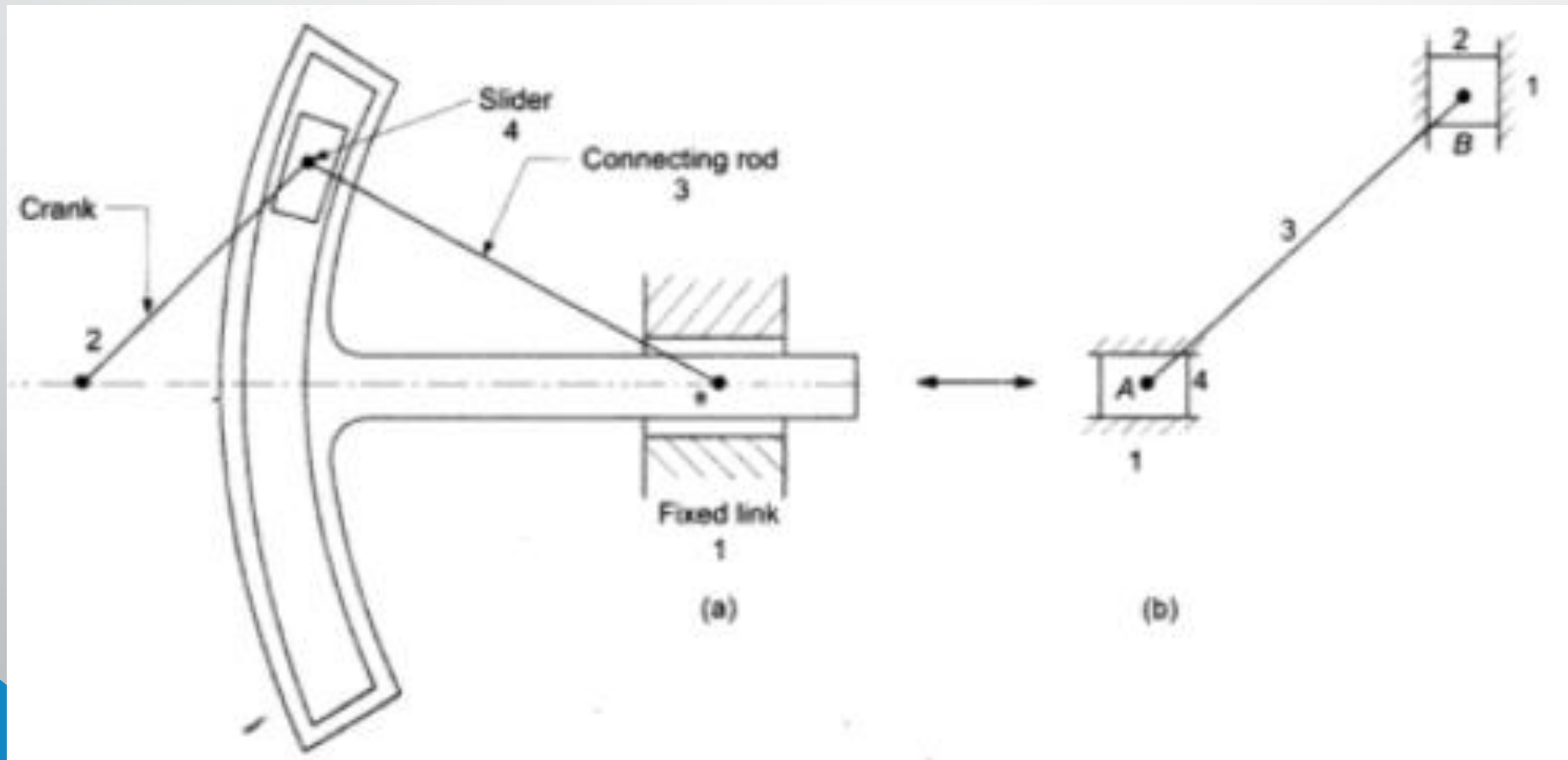
$$F = 2P \tan \alpha$$

- If α approaches zero, for a given F , P approaches infinity.
- A stone crusher utilizes this mechanism to overcome a large resistance with a small force.
- It can be used in numerous toggle clamping devices, for holding work pieces.



Inversions of a Double Slider Crank Chain are:

- A kinematic chain consisting of two turning pairs sliding pairs is called double slider-crank chain.
- Links 3 and 4 reciprocate, link 2 rotates and link 1 is fixed.
- Two pairs of the same kind are adjacent.



Inversions of a Double Slider Crank Chain are:

Donkey Pump:

- Link 2 (crank) rotates about point A.
- One end of the crank is connected to the piston, through the piston rod, which reciprocates vertically in the pump cylinder.
- This cylinder together with the body of the pump represents the fixed link 1.
- The other end of the crank is connected to the slider (link 3) which reciprocates horizontally in the cylinder.

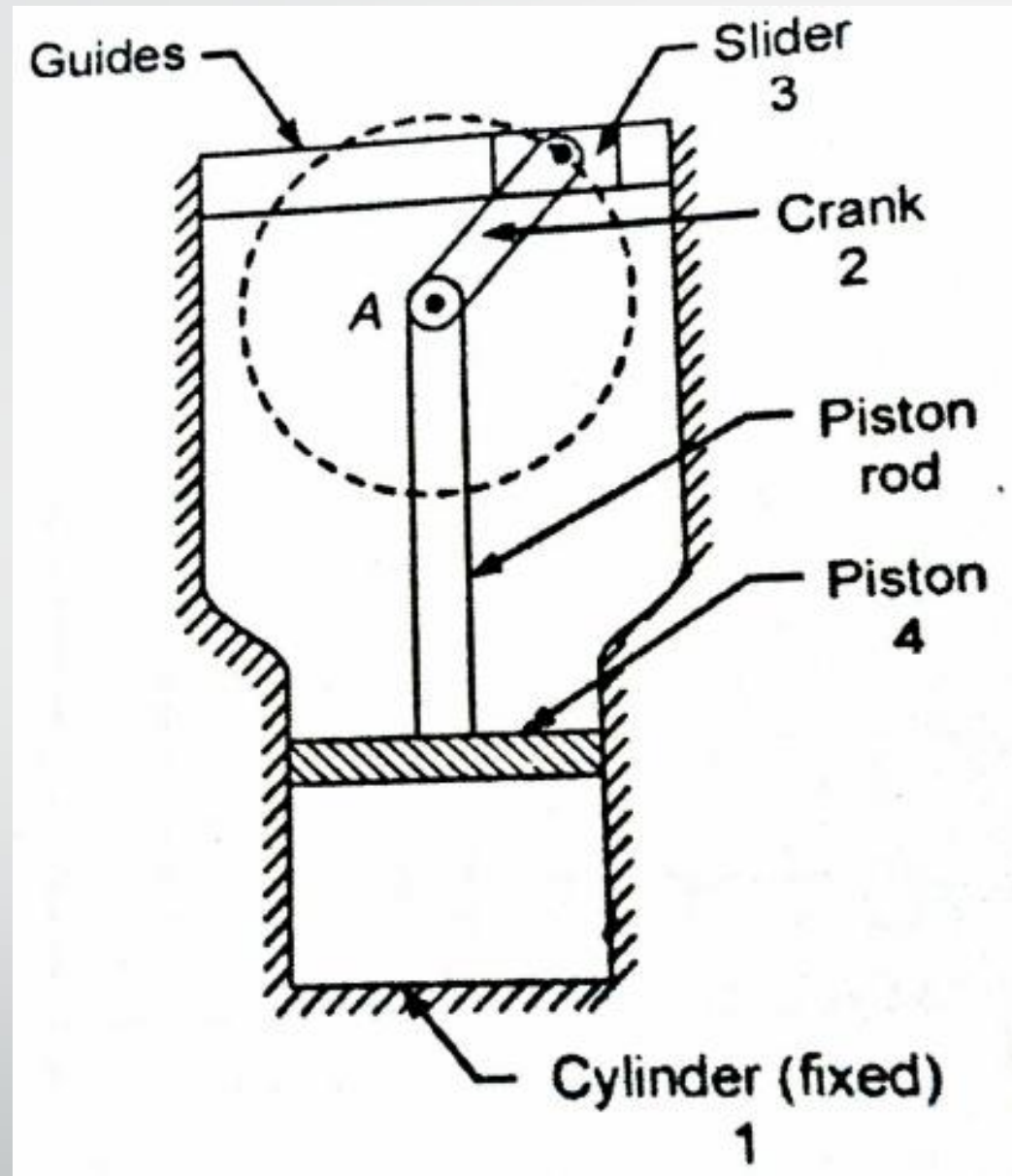
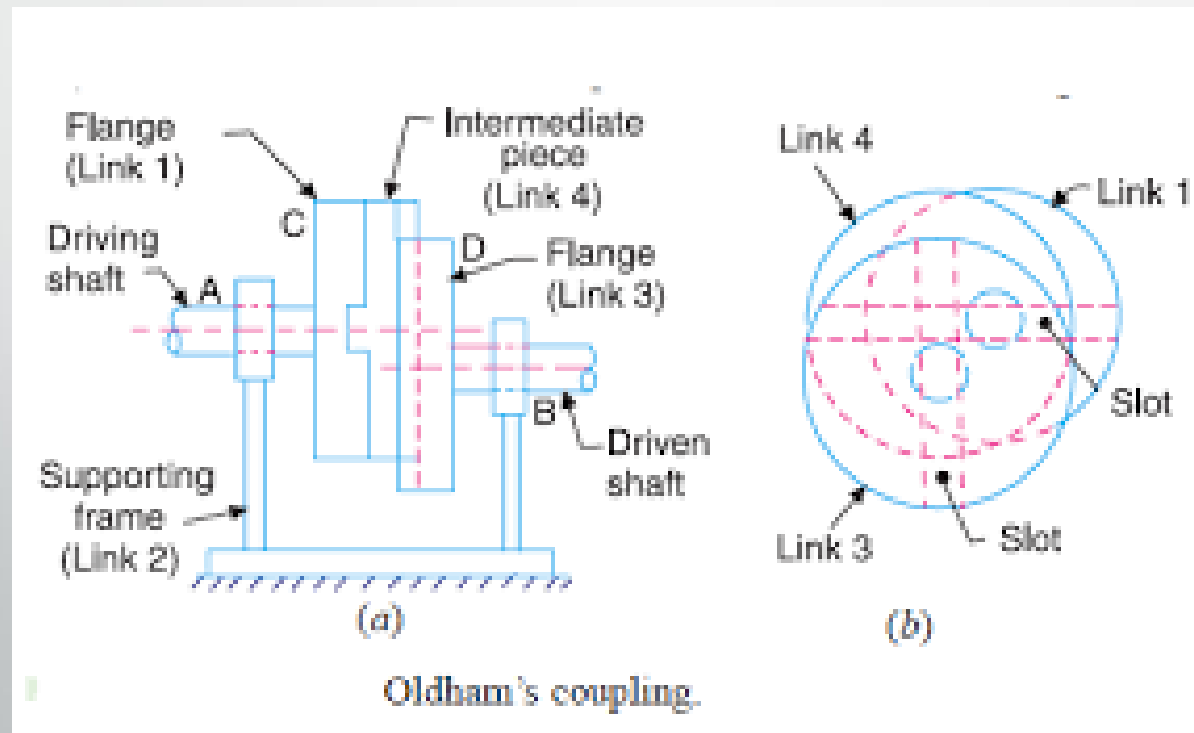


Fig. Donkey Pump

Inversions of a Double Slider Crank Chain are:

Oldham's coupling:

- It is used to connect two parallel shaft, the distance between whose axes is small and variable.
- The shafts connected by the coupling rotate at the same speed.
- The shafts have flange at the ends, in which slots are cut.
- These form a link 1 and 3.



Inversions of a Double Slider Crank Chain are:

Elliptical trammel:

- It is a device to draw ellipse.
- In which two grooves are cut at right angles in a plate that is fixed.
- The plate forms the fixed link 4.
- Two sliding blocks are fitted into the grooves.
- The sliders form two sliding links 1 and 3.
- The link joining slides form the link 2.

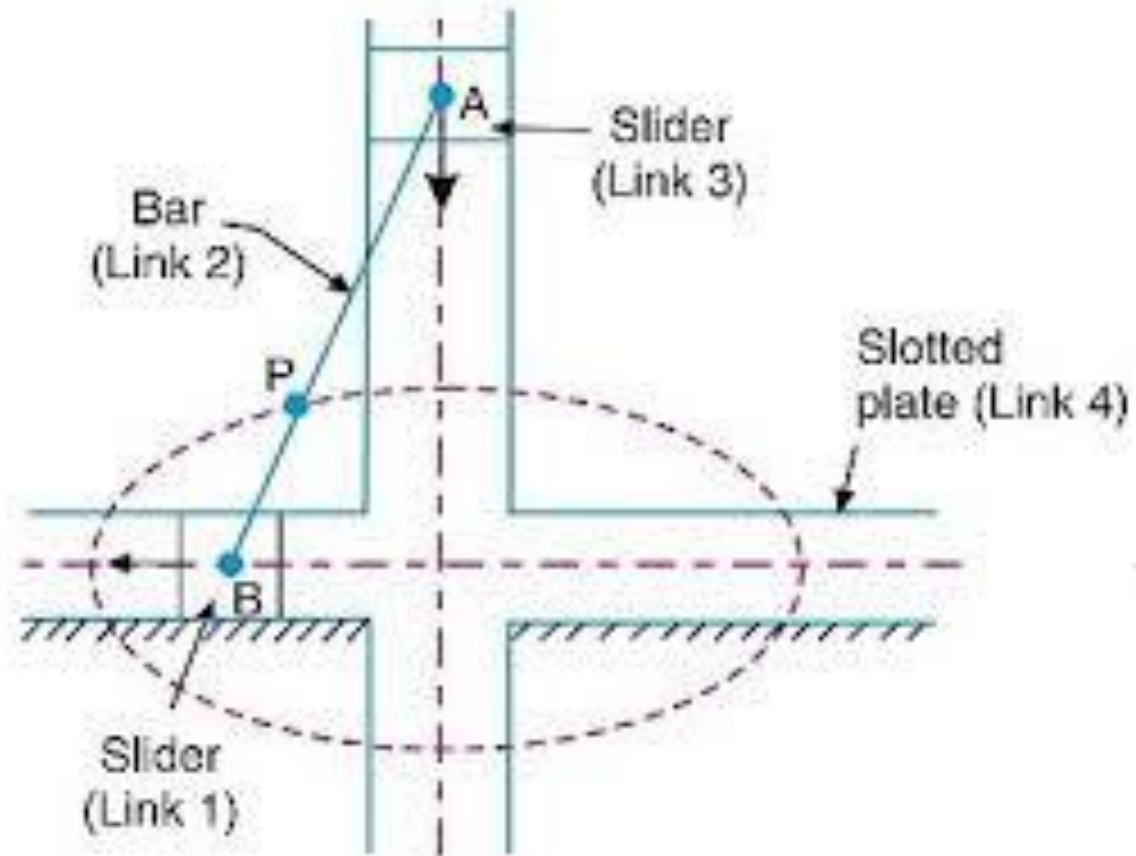
$$y = AB \sin \theta$$

or

$$\frac{y}{AB} = \sin \theta$$

Squaring and adding, we get

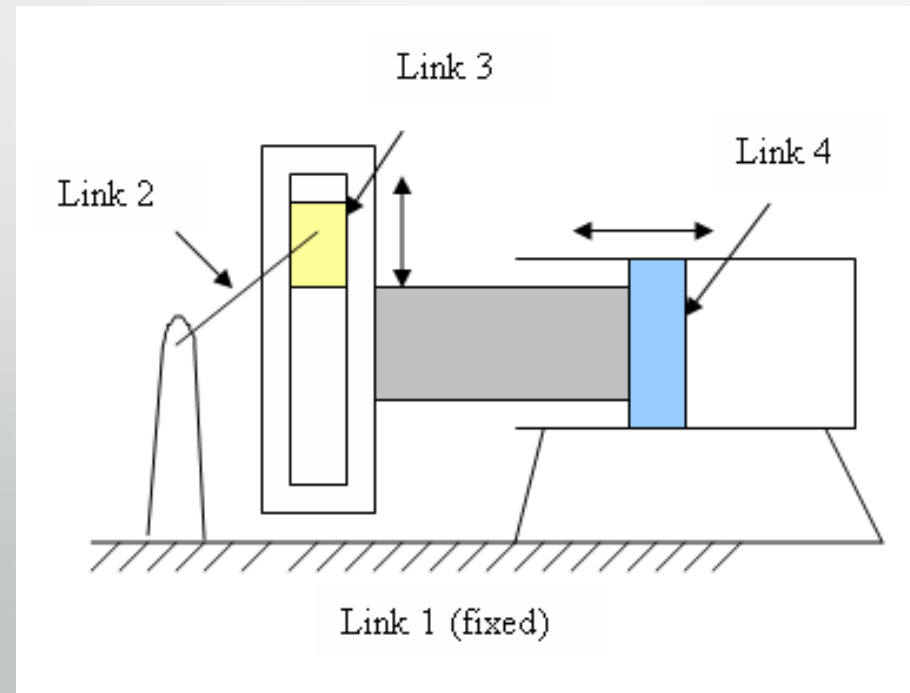
$$\frac{x^2}{BP^2} + \frac{y^2}{AB^2} = 1$$



Inversions of a Double Slider Crank Chain are:

Scotch yoke:

- This mechanism gives simple harmonic motion.
- Its early application was on steam pump.
- Now this mechanism is used for test machine on produce vibrations.
- This is also used as a sine-cosine generator for computing elements.



Example 1: In a crank and slotted lever quick-return mechanism shown in Figure, the distance between the fixed centres is 300 mm and the length of the driving crank is 150 mm. Find the inclination of the slotted lever with the vertical in the extreme position and the ratio of time of cutting stroke to return stroke.

Solution:

Given $AC = 300$ mm, $B_1C = 150$ mm

$$\cos(\beta/2) = B_1C / AC$$

$$= \frac{150}{300} = 0.50$$

$$\beta = 120^\circ$$

$$\alpha = 360^\circ - 120^\circ = 240^\circ$$

$$\frac{\text{Time of cutting stroke}}{\text{Time of return stroke}} = \frac{\alpha}{\beta} = \frac{240^\circ}{120^\circ} = 2$$

Inclination of slotted lever with the vertical $= 90^\circ - \frac{\beta}{2} = 90^\circ - \frac{120^\circ}{2}$
 $= 30^\circ$

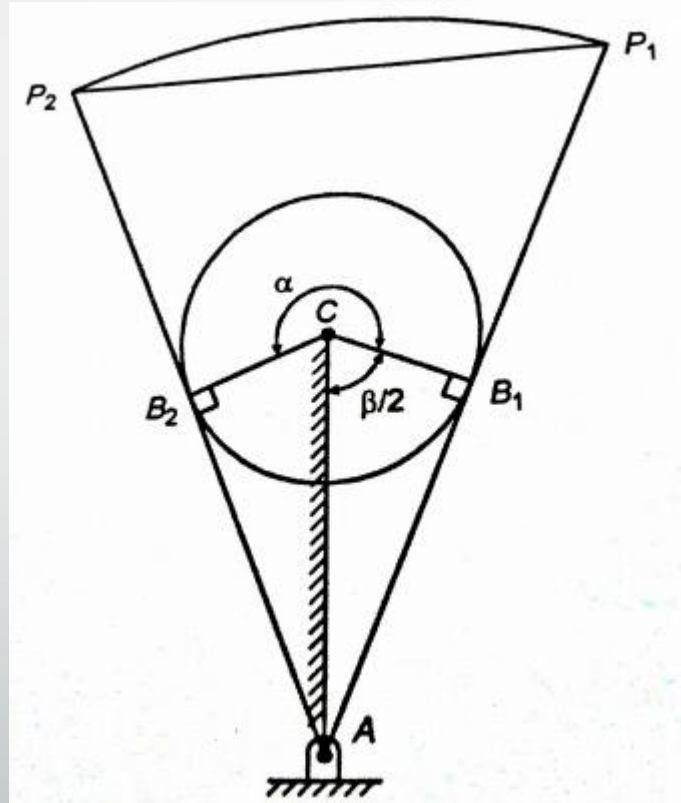


Fig. Crank and Slotted lever mechanism



Thank You